

Base Metals Supply & Demand Tracker

Chinese exports are the relief valve for ex-China aluminium tightness

Copper:

- **China's apparent and end-use copper consumption diverge markedly.** Our consumption-weighted end-use indicator for Chinese copper demand declined by 4% yoy (now +1.2% yoy YTD through 4M26) amid continued weakness in renewables installations and white goods production (Figure 22). The slowdown comes despite continued strong grid demand, EV production and product exports which work to offset weakness in other sectors. At the same time, Chinese apparent consumption was remarkable in April (+9% yoy), with inventory draws boosting overall figures reflecting the level of inventory destock across the supply chain over the 4Q25/1Q26 buyers strike in China. More recently, drawing momentum in Chinese inventory has leveled off in recent weeks as copper prices have ranged between \$13,500-14,000/mt (Figure 23).
- **US tariff review forthcoming.** Later this month, by June 30, 2026, the US Commerce Secretary is to provide the President with an update on whether additional Section 232 copper tariff modifications are necessary. Our view remains that it is more likely than not that the Trump administration will enact an escalating tariff as was laid out in last July's original executive order (i.e. starting at 15% in 2027, increasing to 30% a year later) to 1) ensure that all the copper that has already been imported into the US stays onshore and 2) keep this excess US inventory as a critical reserve, rather than enact policy which incentivizes it to quickly destock in the coming years. A continued strong import pull into the US in the coming quarters would continue to strain ex-US cathode supply, particularly over periods where China's import arb is also open.

Aluminium:

- **Chinese aluminium semis and finished product exports are the relief valve for ex-China tightness.** As LME inventories continue to draw lower to 330 kmt, Chinese visible inventory levels are still sitting near multi-year highs at ~1.4 mmt. With the semis export arb from China to rest of Asia open and at attractive levels (Figure 72), we continue to expect strong exports out of China to accelerate weekly withdrawals from onshore aluminium inventory in China (Figure 53 & Figure 55). So far, in the last 6 weeks only around 90 kmt has been drawn from SHFE and regional warehouses in China as production levels have risen and imports of aluminum have remained firm, with drawing activity only picking up since mid-May.
- **Producers in China are reaping the rewards of higher aluminium prices.** Refined production in China in April grew by 3.4% yoy, running at an annualized level of 45 mmt, contributing to YTD growth of ~4% yoy according to CRU (Figure 70). Chinese producers are currently highly incentivized to keep churning out production towards the limiting cap amid elevated aluminium prices. We continue to monitor for [disruptions of bauxite exports](#) from Guinea, as this could become a serious bottleneck for Chinese producers

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as 75% of the country's bauxite imports come from Guinea alone. While bauxite imports from Guinea were up by 13% yoy in April, a sharp change here could eventually threaten Chinese output and potentially also influence Chinese policy around exports as domestic supply security becomes more critical.

Zinc:

- **Chinese concentrate imports cooled in April though refined output stayed strong.** China's zinc concentrate imports fell by nearly 10% yoy in April, as a lack of concentrate availability is beginning to emerge with spot TCs in China flipping to negative territory since the end of March. Smelter production, on the other hand, remained robust increasing by 5% yoy in April, as smelters work through concentrate inventory following strong imports in 1Q26. Even tighter concentrate markets could eventually weigh on Chinese output, which remains a risk for a tighter than expected refined balance this year.
- **Subdued demand continues to show via elevated and stagnant visible inventory.** Total SHFE + social warehouse inventories in China have stabilized near 264 kmt as galvaniser operating rates remain subdued and construction activity remains lifeless in China (Figure 76). Galvaniser demand continues to remain below the 5-yr average, while apparent demand is down by 12% yoy as of April (Figure 78 & Figure 86).

Nickel:

- **Visible inventory builds stagnate.** While inventory on SHFE continues to rise, LME stocks modestly declined from a peak in February, leading to net inventory growth of 70 kmt YTD to a total inventory level of ~470 kmt. Meanwhile, Indonesia's policy framework is set to weigh on producer margins amid higher ore benchmarks, potentially weighing on refined output somewhat this year, though the market remains focused on any upcoming additions to Indonesia's ore quota into the summer.

China demand indicators:

- **China's grid investment remains solid as renewables remain under pressure.** Following a sharp jump at the start of the year, April grid spend rose by 9% yoy, resulting in >30% growth yoy on a ytd basis (Figure 5). However, renewable installations continue to stumble, coming off the rapid pace of installations in 1H2025. Solar installations in April decelerated to ~9.5GW in April, decreasing by 79% yoy, while wind installations declined by 7% yoy to 5.4GW (Figure 7 & Figure 8).
- **NEV production in China rebounded in April.** Total EV and PHEV production through April in China rose by 4% yoy while EV sales increased by 2.5% yoy in April. US EV sales continued to struggle while other regions such as Europe saw healthy EV sales. Overall, global passenger vehicle sales decreased by 10% yoy in April, with China and US taking the biggest hit (Figure 1).
- **White goods demand stumbled in April.** Our estimate for total aggregate white goods copper demand in China decreased by 3% yoy in April, driven by a 9% decline in AC production.

Global demand indicators

Table 1: Global metals monthly price change heatmap

Percent change, MoM & YoY

Global Metals Price Heatmap		May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26	May-26	2023	2024	2025	2026 YTD
Metals Prices	LME Copper	4.1%	3.9%	-2.6%	3.0%	3.7%	6.0%	2.8%	11.0%	5.9%	1.4%	-9.3%	5.3%	5.0%	2.2%	2.4%	41.7%	10.5%
	LME Aluminium	1.9%	6.3%	-1.3%	2.0%	2.5%	7.6%	-0.6%	4.4%	5.0%	-0.1%	3.8%	0.2%	5.5%	0.3%	7.0%	17.4%	21.7%
	LME Zinc	1.1%	5.0%	0.3%	2.1%	5.0%	3.2%	0.0%	2.0%	9.1%	-2.5%	-8.3%	4.2%	5.3%	-10.6%	12.1%	4.7%	14.5%
	LME Nickel	-1.2%	-0.1%	-1.8%	3.2%	-1.2%	-0.1%	-2.6%	12.3%	7.9%	-0.6%	-5.0%	13.8%	-2.1%	-44.7%	-7.7%	8.6%	11.3%
	LME Lead	0.0%	4.4%	-3.6%	1.0%	-0.1%	1.4%	-1.8%	1.5%	-0.1%	-2.3%	-3.5%	2.8%	3.1%	-9.8%	-5.6%	3.0%	-0.1%
	LME Tin	-3.0%	10.9%	-3.0%	7.1%	1.1%	1.9%	8.5%	3.6%	28.1%	11.1%	-23.4%	5.3%	12.6%	2.4%	14.4%	39.4%	33.4%
	LME Cobalt spot	0.0%	-1.1%	0.0%	0.0%	5.1%	39.3%	0.0%	9.9%	5.5%	0.1%	0.0%	0.0%	0.0%	-44.3%	-15.4%	117.8%	5.6%
	LME Lithium Hydroxide	-6.9%	-3.2%	-2.1%	5.4%	11.9%	1.0%	5.8%	10.1%	62.0%	-4.3%	12.5%	3.5%	7.8%	-80.4%	-42.9%	18.5%	83.4%
	COMEX Copper	2.6%	7.5%	-13.4%	3.8%	7.5%	4.8%	1.9%	9.6%	4.3%	1.4%	-9.7%	5.6%	7.8%	2.1%	3.5%	41.1%	12.8%
	SHFE Copper	0.5%	2.4%	-1.9%	1.0%	5.1%	4.8%	0.0%	13.1%	8.3%	-4.2%	-7.8%	5.8%	3.6%	4.4%	6.8%	33.7%	6.8%
	SHFE Aluminium	0.6%	2.8%	-0.5%	0.6%	0.0%	2.5%	1.1%	5.6%	10.6%	-6.0%	0.1%	-1.3%	-0.5%	4.5%	1.0%	14.7%	6.8%
	SHFE Zinc	-0.7%	-0.5%	-0.6%	-1.3%	-1.0%	2.0%	0.2%	4.1%	12.2%	-6.0%	-6.3%	0.4%	5.1%	-9.5%	19.0%	-9.4%	6.3%
	Spot Gold	0.0%	0.4%	-0.4%	4.8%	11.9%	3.7%	5.9%	1.9%	13.3%	7.9%	-15.2%	-1.1%	-1.7%	13.1%	27.2%	64.6%	3.3%
	Spot Silver	1.1%	9.5%	1.7%	8.2%	17.4%	4.4%	16.0%	26.8%	18.9%	10.1%	-24.1%	-1.9%	2.1%	-0.7%	21.5%	148.0%	1.1%
	Spot Platinum	9.1%	28.5%	-5.0%	6.1%	14.9%	-0.1%	6.1%	23.3%	6.5%	7.9%	-18.5%	1.7%	-3.4%	-7.7%	-8.5%	127.0%	-8.7%
	Spot Palladium	3.2%	13.6%	8.3%	-7.9%	14.3%	14.1%	1.1%	11.4%	5.7%	4.4%	-19.4%	3.6%	-11.2%	-38.6%	-17.1%	77.5%	-19.6%

*LME is 3-month rolling forward, except cobalt which is spot. Lithium hydroxide and SHFE prices are futures contracts. Prices as of 5th June 2026.

Source: Bloomberg Finance L.P.

Table 2: Global manufacturing PMI by country

		Jan-25	Feb-25	Mar-25	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26
Groups	JP Morgan Global	50.1	50.6	50.3	49.8	49.5	50.4	49.7	50.9	50.7	50.7	50.5	50.4	50.9	51.8	51.3	52.6
	Developed Markets	49.3	50.0	49.1	49.1	50.0	50.5	49.2	50.9	50.3	50.3	50.5	50.5	51.4	51.8	52.0	53.8
	Emerging Markets	50.8	51.1	51.3	50.5	49.2	50.4	50.1	50.9	51.2	51.2	50.5	50.4	50.6	52.0	50.7	51.6
	European Union	46.7	47.8	48.6	49.1	49.2	49.2	49.6	50.4	49.7	49.7	49.5	48.8	49.5	50.5	51.3	52.0
	Eurozone	46.6	47.6	48.6	49.0	49.4	49.5	49.8	50.7	49.8	49.8	49.6	48.8	49.5	50.8	51.6	52.2
Asia Pacific	China (Caixin/Markit)	50.1	50.8	51.2	50.4	48.3	50.4	49.5	50.5	51.2	51.2	49.9	50.1	50.3	52.1	50.8	52.2
	China (Official PMI)	49.1	50.2	50.5	49.0	49.5	49.7	49.3	49.4	49.8	49.8	49.2	50.1	49.3	49.0	50.4	50.3
	India	57.7	56.3	58.1	58.2	57.6	58.4	59.1	59.3	57.7	57.7	56.6	55.0	55.4	56.9	53.9	54.7
	Japan	48.7	49.0	48.4	48.7	49.4	50.1	48.9	49.7	48.5	48.5	48.7	50.0	51.5	53.0	51.6	55.1
	France	45.0	45.8	48.5	48.7	49.8	48.1	48.2	50.4	48.2	48.2	47.8	50.7	51.2	50.1	50.0	52.8
Europe	Germany	45.0	46.5	48.3	48.4	48.3	49.0	49.1	49.8	49.5	49.5	48.2	47.0	49.1	50.9	52.2	51.4
	Italy	46.3	47.4	46.6	49.3	49.2	48.4	49.8	50.4	49.0	49.0	50.6	47.9	48.1	50.6	51.3	52.1
	Spain	50.9	49.7	49.5	48.1	50.5	51.4	51.9	54.3	51.5	51.5	51.5	49.6	49.2	50.0	48.7	51.7
	United Kingdom	48.3	46.9	44.9	45.4	46.4	47.7	48.0	47.0	46.2	46.2	50.2	50.6	51.8	51.7	51.0	53.7
	Canada	51.6	47.8	46.3	45.3	46.1	45.6	46.1	48.3	47.7	47.7	48.4	48.6	50.4	51.0	50.0	53.3
Americas	Mexico	49.1	47.6	46.5	44.8	46.7	46.3	49.1	50.2	49.6	49.6	47.3	46.1	46.3	47.1	48.9	47.7
	United States	51.2	52.7	50.2	50.2	52.0	52.9	49.8	53.0	52.0	52.0	52.2	51.8	52.4	51.6	52.3	54.5
	US ISM	50.5	50.0	48.9	48.8	48.6	49.0	48.4	48.9	48.9	48.9	48.0	47.9	52.6	52.4	52.7	52.7

Source: Bloomberg Finance L.P.

Table 3: Global light vehicle production growth

Percent change

		1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26	4Q26	1Q27	2Q27	3Q27	4Q27	2024	2025	2026E	2027F
QoQ	Europe	0.5%	2.4%	-13.4%	12.0%	0.8%	-1.6%	-11.9%	11.0%	0.0%	4.2%	-12.6%	11%				
	China	-24.5%	8.0%	7.8%	17.1%	-32.1%	13.9%	10.4%	14.6%	-26.4%	16.8%	-8.6%	26%				
	Japan/Korea	-5.4%	-0.4%	-3.3%	6.0%	0.0%	-4.1%	-6.4%	2.0%	-0.5%	1.0%	-3.5%	6%				
	North America	4.2%	5.0%	0.3%	-9.9%	4.1%	4.0%	0.0%	-9.9%	4.0%	6.1%	-2.2%	-1%				
	Other	4.4%	-4.9%	6.9%	1.1%	0.1%	-7.2%	4.6%	-3.4%	2.4%	4.6%	5.4%	-2%				
	Global	-8.9%	3.0%	0.8%	7.4%	-11.9%	2.7%	1.2%	5.2%	-9.7%	8.3%	-5.3%	11%				
	YoY																
YoY	Europe	-5.3%	-0.3%	3.8%	-0.2%	0.2%	-3.8%	-2.0%	-2.9%	-3.7%	2.0%	1.1%	0.9%	-4.4%	-0.7%	-2.1%	0.0%
	China	14.5%	11.0%	14.6%	2.8%	-7.5%	-2.4%	-0.1%	-2.2%	6.1%	8.8%	-10.0%	-0.7%	3.6%	10.0%	-2.9%	0.5%
	Japan/Korea	7.0%	0.3%	0.8%	-3.5%	2.1%	-1.7%	-4.8%	-8.4%	-8.9%	-4.1%	-1.2%	2.7%	-6.4%	1.0%	-3.2%	-3.0%
	North America	-5.0%	-3.5%	5.2%	-1.1%	-1.2%	-2.2%	-2.5%	-2.5%	-2.6%	-0.6%	-2.7%	7.0%	-1.5%	-1.2%	-2.1%	0.1%
	Other	4.3%	5.6%	1.4%	7.3%	2.9%	0.4%	-1.8%	-6.2%	-4.1%	8.1%	9.0%	10.1%	-0.5%	4.6%	-1.2%	5.6%
	Global	3.9%	3.7%	6.9%	1.6%	-1.8%	-2.1%	-1.7%	-3.8%	-1.3%	4.1%	-2.6%	2.8%	-1.0%	3.9%	-2.4%	0.8%

*2Q26-4Q27 is a forecast as of May 2026.

Source: S&P Global

Figure 1: Vehicle sales and penetration rates

Vehicles in thousands

Global PV Sales ('000s)	Apr-26	Apr-25	y/y%	4M26	4M25	y/y%
Passenger Vehicles (PV)						
Europe (EU5)	799	739	8.0%	3,302	3,118	5.9%
Germany	249	243	2.7%	949	907	4.5%
France	138	139	-0.3%	540	549	-1.6%
Italy	155	139	11.6%	642	583	10.1%
Spain	107	99	8.5%	407	378	7.8%
United Kingdom	149	120	24.0%	764	701	9.0%
China	1,384	1,755	-21.1%	5,609	6,872	-18.4%
US	1,380	1,466	-5.9%	5,096	5,426	-6.1%
Total	3,563	3,960	-10.0%	14,006	15,416	-9.1%
Battery Electric Vehicles (EV)						
Europe (EU5)	163	109	48.9%	637	459	38.8%
Germany	64	46	41.3%	224	159	41.3%
France	36	26	41.8%	148	100	48.2%
Italy	13	7	97.6%	51	30	73.1%
Spain	10	7	42.3%	37	26	41.8%
United Kingdom	39	25	59.1%	177	145	22.1%
China	579	565	2.5%	1,784	2,068	-13.7%
US	78	102	-23.4%	298	410	-27.1%
Total	820	776	5.6%	2,720	2,937	-7.4%
Plug-in Hybrid Vehicles (PHEV)						
Europe (EU5)	83	65	28.6%	327	241	35.7%
Germany	28	24	13.3%	104	88	17.5%
France	8	10	-13.1%	22	29	-25.9%
Italy	14	8	79.3%	54	26	104.9%
Spain	13	9	42.9%	49	30	64.4%
United Kingdom	21	14	46.4%	99	68	46.5%
China	271	346	-21.7%	978	1,264	-22.6%
US	15	32	-53.3%	49	127	-61.0%
Total	370	443	-16.6%	1,355	1,632	-17.0%

Global EV/PHEV Penetration (%)	Apr-26	Apr-25	y/y%	4M26	4M25	y/y%
EV/PHEV Penetration (%)						
Europe (EU5)	30.8%	23.5%	7.3%p	29.2%	22.5%	6.8%p
Germany	36.9%	28.8%	8.1%p	34.5%	27.2%	7.3%p
France	32.2%	25.3%	6.9%p	31.5%	23.6%	7.9%p
Italy	17.5%	10.4%	7.1%p	16.4%	9.6%	6.8%p
Spain	21.3%	16.2%	5.1%p	21.0%	14.7%	6.3%p
United Kingdom	40.0%	32.1%	7.9%p	36.1%	30.3%	5.8%p
China	61.4%	51.9%	9.5%p	49.2%	48.5%	0.8%p
US	6.8%	9.2%	-2.4%p	6.8%	9.9%	-3.1%p
Total	33.4%	30.8%	2.6%p	29.1%	29.6%	-0.5%p
EV Penetration (%)						
Europe (EU5)	20.4%	14.8%	5.6%p	19.3%	14.7%	4.6%p
Germany	25.8%	18.8%	7.1%p	23.6%	17.5%	6.1%p
France	26.2%	18.4%	7.8%p	27.5%	18.2%	9.2%p
Italy	8.5%	4.8%	3.7%p	8.0%	5.1%	2.9%p
Spain	9.1%	6.9%	2.2%p	9.1%	6.9%	2.2%p
United Kingdom	26.2%	20.4%	5.8%p	23.1%	20.7%	2.5%p
China	41.8%	32.2%	9.6%p	31.8%	30.1%	1.7%p
US	5.7%	7.0%	-1.3%p	5.9%	7.5%	-1.7%p
Total	23.0%	19.6%	3.4%p	19.4%	19.0%	0.4%p
PHEV Penetration (%)						
Europe (EU5)	10.5%	8.8%	1.7%p	9.9%	7.7%	2.2%p
Germany	11.1%	10.0%	1.0%p	10.9%	9.7%	1.2%p
France	6.0%	6.9%	-0.9%p	4.0%	5.3%	-1.3%p
Italy	9.0%	5.6%	3.4%p	8.4%	4.5%	3.9%p
Spain	12.2%	9.3%	2.9%p	12.0%	7.8%	4.1%p
United Kingdom	13.8%	11.7%	2.1%p	13.0%	9.7%	3.3%p
China	19.6%	19.7%	-0.1%p	17.4%	18.4%	-1.0%p
US	1.1%	2.2%	-1.1%p	1.0%	2.3%	-1.4%p
Total	10.4%	11.2%	-0.8%p	9.7%	10.6%	-0.9%p

*As of April 2026.

Source: CNEV Post, European Union, US Motor Intelligence, J.P. Morgan. Note: China data are based on preliminary CPCA retail sales data.

China demand indicators

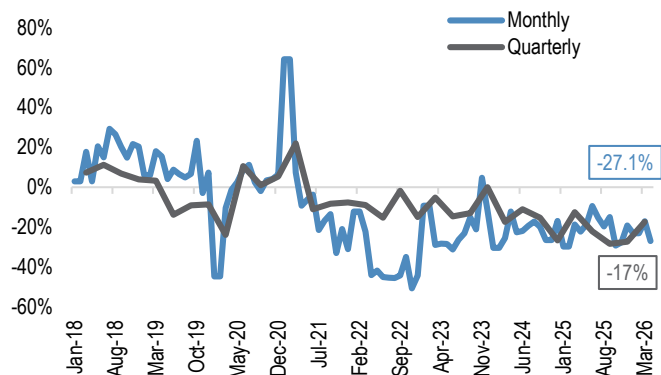
Table 4: China metals demand indicator heat map

China End-Use Demand Indicators Heatmap		Units	Jan-25	Feb-25	Mar-25	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26
Property	70 cities newly built price index	YoY	-5.4%	-5.2%	-5.0%	-4.6%	-4.1%	-3.7%	-3.4%	-3.0%	-2.7%	-2.6%	-2.8%	-3.1%	-3.3%	-3.5%	-3.6%	-3.7%
	Residential building starts (million sqm)	YTD YoY	-28.9%	-28.9%	-23.9%	-22.3%	-21.4%	-19.6%	-18.3%	-18.3%	-18.3%	-19.3%	-19.9%	-19.8%	-23.3%	-23.3%	-22.0%	-23.6%
	Residential area under const. (million sqm)	YTD YoY	-9.7%	-9.7%	-9.9%	-10.1%	-9.6%	-9.4%	-9.4%	-9.6%	-9.7%	-9.7%	-10.0%	-10.3%	-11.9%	-11.9%	-12.1%	-12.5%
	Residential const. completed (m. sqm)	YTD YoY	-17.7%	-17.7%	-14.7%	-16.8%	-17.6%	-15.5%	-17.3%	-18.5%	-17.1%	-18.9%	-20.1%	-20.2%	-26.9%	-26.9%	-26.5%	-25.8%
Infrastructure	FAI - total (RMB)	YTD YoY	4.1%	4.1%	4.2%	4.0%	3.7%	2.8%	1.6%	0.5%	-0.5%	-1.7%	-2.6%	-3.8%	1.8%	1.8%	1.7%	-1.6%
	FAI- manufacturing (RMB)	YTD YoY	9.0%	9.0%	9.1%	8.8%	8.5%	7.5%	6.2%	5.1%	4.0%	2.7%	1.9%	0.6%	3.1%	3.1%	4.1%	1.2%
	FAI- electric equipment and machinery (RMB)	YTD YoY	-8.6%	-8.6%	-7.4%	-7.5%	-8.6%	-7.8%	-8.7%	-8.8%	-9.5%	-9.4%	-9.5%	-10.3%	5.8%	5.8%	0.8%	2.9%
	FAI- railway transportation (RMB)	YTD YoY	0.2%	0.2%	0.5%	1.6%	2.3%	4.2%	5.9%	4.5%	4.2%	3.0%	2.7%	-1.2%	28.7%	28.7%	12.6%	8.6%
	FAI- construction (RMB)	YTD YoY	-32.6%	-32.6%	-12.6%	-0.3%	-19.8%	-21.2%	-24.7%	-24.7%	-17.2%	-13.7%	-14.7%	-22.2%	2.2%	2.2%	0.1%	-11.7%
	FAI- real estate (RMB)	YTD YoY	-9.1%	-9.1%	-10.0%	-10.6%	-11.1%	-11.5%	-12.4%	-13.2%	-14.0%	-14.9%	-16.0%	-17.5%	-10.7%	-10.7%	-10.6%	-13.4%
	FAI- investment in grid infrastructure (RMB)	YTD YoY	33.3%	33.3%	24.8%	14.6%	19.8%	14.6%	12.5%	14.0%	9.9%	7.2%	5.9%	5.1%	92.1%	92.1%	43.4%	32.4%
Manufacturing	Cement production (million tonnes)	YTD YoY	-5.7%	-5.7%	2.5%	-5.3%	-8.1%	-5.3%	-5.6%	-6.2%	-8.6%	-15.8%	-8.2%	-6.6%	6.8%	6.8%	-21.0%	-10.8%
	China NBS Manufacturing PMI	Index	49.1	50.2	50.5	49.0	49.5	49.7	49.3	49.4	49.8	49.0	49.2	50.1	49.3	49.0	50.4	50.3
	Refrigerator Production (units)	YTD YoY	10.0%	10.0%	2.2%	-2.0%	-2.6%	0.3%	1.7%	3.5%	3.8%	3.1%	3.6%	5.1%	10.1%	10.1%	12.7%	15.9%
	Washing Machine Production (units)	YTD YoY	14.1%	14.1%	15.5%	10.5%	9.3%	10.3%	10.9%	9.5%	9.2%	8.2%	8.1%	6.6%	0.3%	0.3%	3.6%	3.9%
	Airconditioner Production (units)	YTD YoY	9.7%	9.7%	8.3%	6.2%	4.7%	4.0%	3.9%	4.3%	3.0%	2.5%	1.2%	0.4%	-2.8%	-2.8%	0.2%	-2.4%
	Freezer Production (units)	YTD YoY	9.3%	-9.3%	-8.6%	-7.0%	-5.7%	-2.1%	0.5%	2.3%	1.9%	-0.7%	0.0%	0.8%	20.5%	20.5%	14.5%	16.2%
	Total Automotive Production (units)	YoY	1.7%	39.6%	11.9%	8.9%	11.6%	11.4%	13.3%	13.0%	17.1%	12.1%	2.8%	-2.1%	0.01%	-20.5%	-3.0%	-1.7%
	Automotive Sales for Passenger Vehicles (units)	YoY	0.8%	36.2%	10.4%	11.0%	13.3%	14.5%	14.7%	16.5%	13.2%	7.5%	1.2%	-8.7%	-6.8%	-15.4%	-2.3%	-4.2%
	Automotive Sales for Commercial Vehicles (units)	YoY	-10.3%	25.0%	-2.4%	2.7%	-2.0%	9.5%	14.1%	16.3%	29.6%	21.0%	24.4%	15.3%	23.5%	-14.0%	8.9%	8.1%
	Automotive Sales for Heavy-Duty Truck (units)	YoY	-20.3%	5.7%	9.3%	37.3%	43.3%	42.3%	83.5%	53.5%	71.5%	55.4%	60.6%	47.9%	47.7%	14.5%	23.3%	16.3%
	Automotive Production for Heavy-Duty Truck (units)	YoY	-18%	37%	7%	31%	40%	40%	76%	64%	80%	96%	69%	70%	20%	23%	23%	23%
	Sales of BEV Passenger Vehicles	YoY	28.8%	96.0%	53.7%	60.5%	45.9%	39.5%	39.7%	37.4%	33.1%	29.8%	23.6%	14.0%	0.4%	-17.6%	-4.2%	-0.8%

Source: Bloomberg Finance L.P., NBS, J.P. Morgan Commodities Research

Figure 2: Building new starts growth

Percent change, yoy

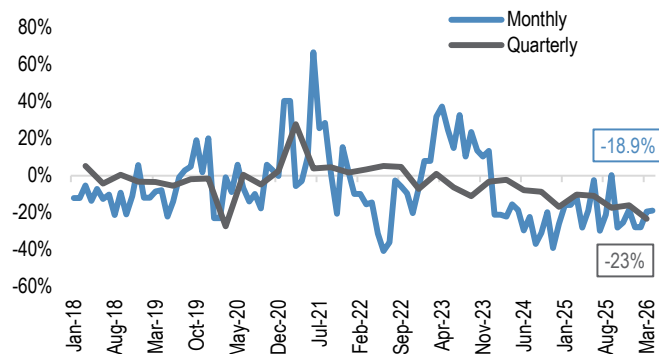


*Includes residential and non-residential. Text box highlights the latest data point; monthly is through April 2026 and quarterly is as of 1Q26.

Source: China NBS

Figure 3: Building completions growth

Percent change, yoy

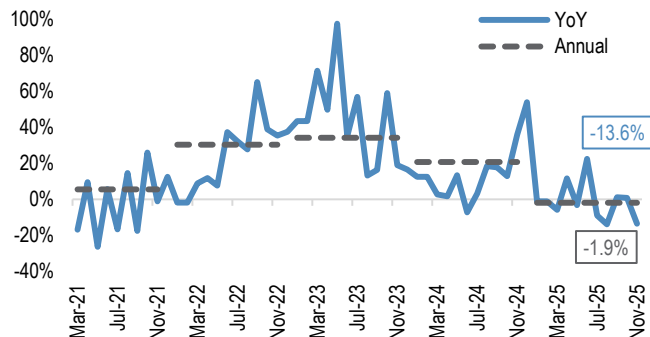


*Includes residential and non-residential. Text box highlights the latest data point; monthly is through April 2026 and quarterly is as of 1Q26.

Source: China NBS

Figure 4: Investment in power generation

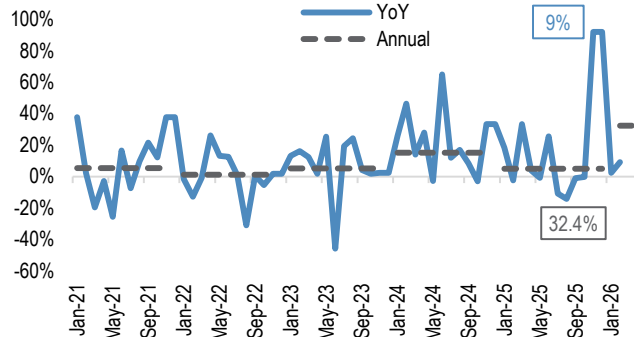
Percent change, yoy



*2025 is through November (lagged).
Source: China National Energy Administration

Figure 5: Fixed asset investment in grid infrastructure

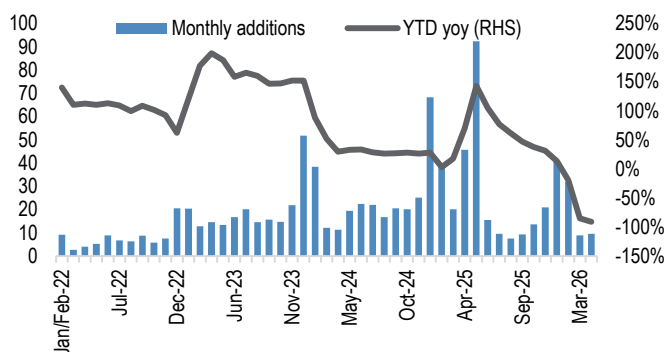
Percent change, yoy



*2026 is through April.
Source: China National Energy Administration. J.P. Morgan estimates for 2M26 based on State Grid and Southern Grid spending.

Figure 6: Solar power plant capacity additions

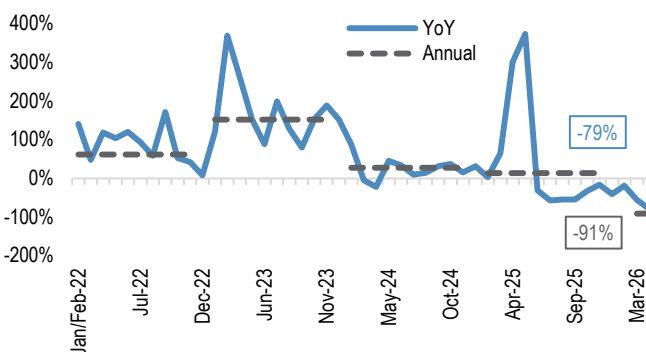
LHS: Gigawatts; RHS: Percent change, YTD yoy



*2026 is through April.
Source: China National Energy Administration

Figure 7: Solar power plant capacity addition growth

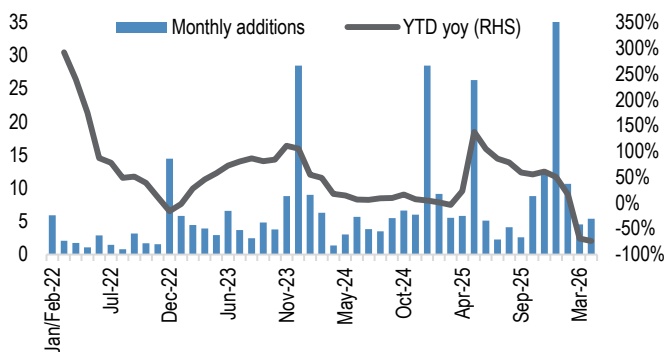
Percent change, yoy



*2026 is through April.
Source: China National Energy Administration

Figure 8: Wind power plant capacity additions

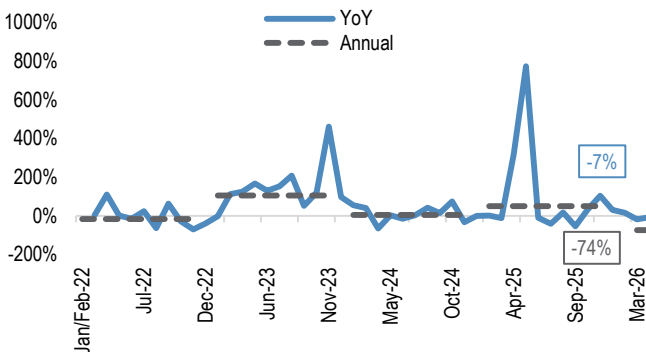
LHS: Gigawatts; RHS: Percent change, yoy



*2026 is through April.
Source: China National Energy Administration

Figure 9: Wind power plant capacity addition growth

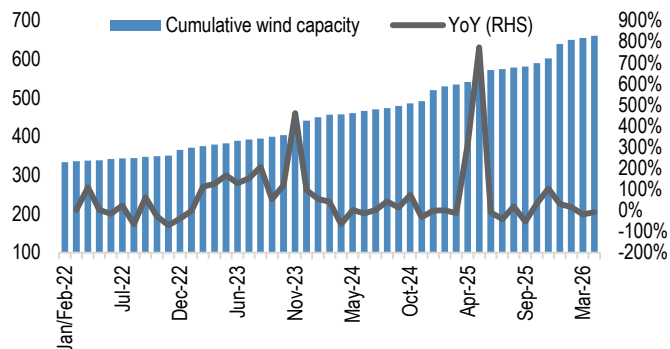
Percent change, yoy



*2026 is through April.
Source: China National Energy Administration

Figure 10: Cumulative wind installed capacity

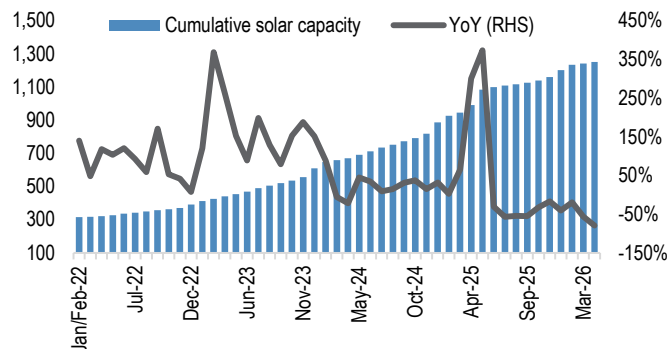
LHS: Gigawatts; RHS: Percent change, yoy of monthly installations



*2026 is through April.
Source: China National Energy Administration

Figure 11: Cumulative solar installed capacity

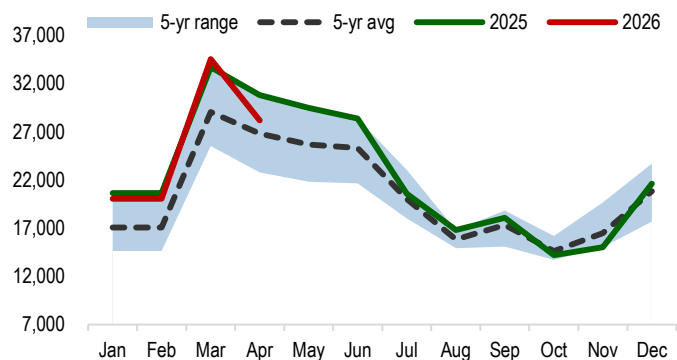
LHS: Gigawatts; RHS: Percent change, yoy of monthly installations



*2026 is through April.
Source: China National Energy Administration

Figure 12: Air conditioner production, monthly

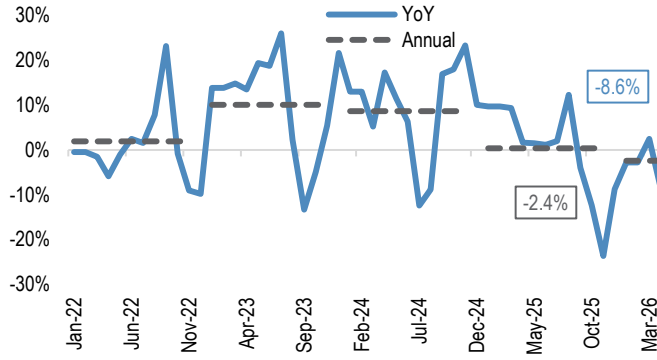
Thousand units



Source: China NBS

Figure 13: Air conditioner production growth

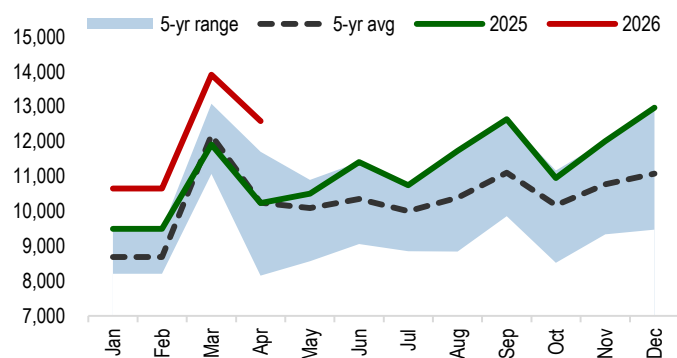
Percent change, yoy



*2026 is through April.
Source: China NBS

Figure 14: Fridge and freezer production, monthly

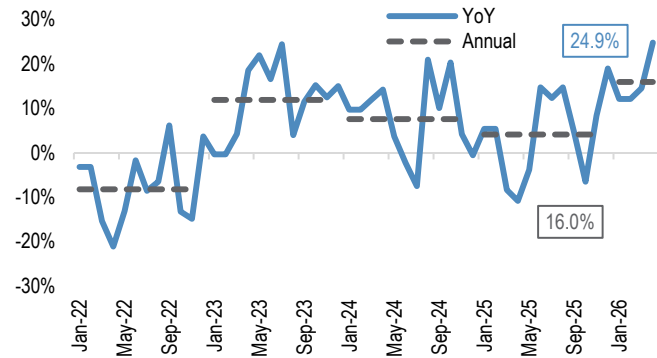
Thousand units



Source: China NBS

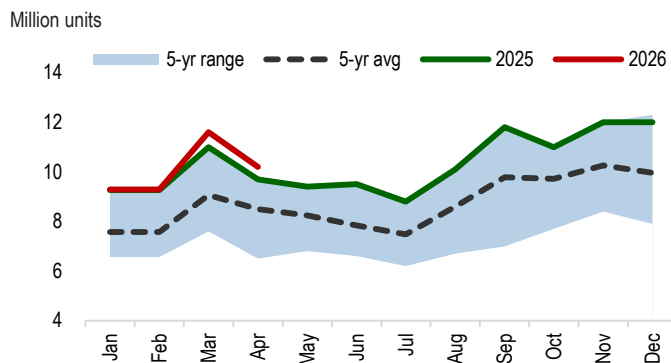
Figure 15: Fridge and freezer production growth

Percent change, yoy



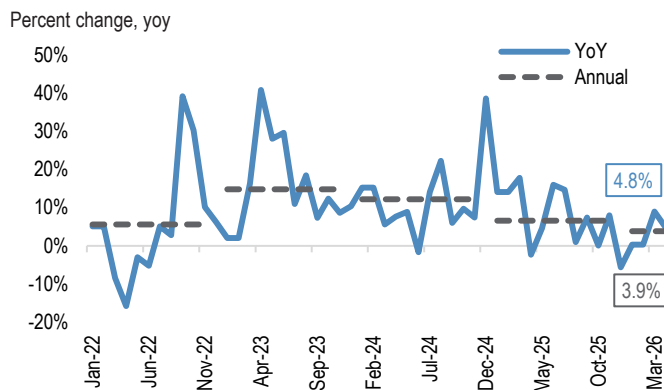
*2026 is through April.
Source: China NBS

Figure 16: Washing machine production, monthly



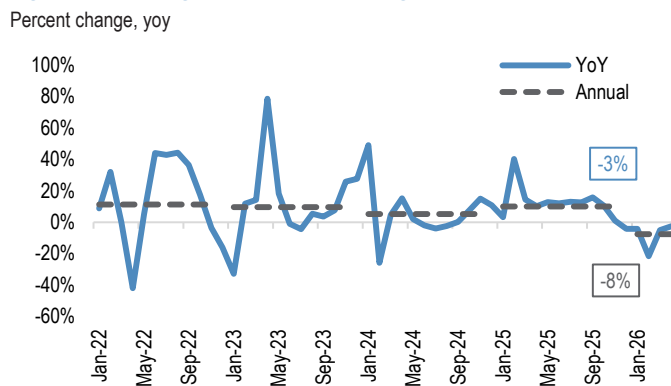
Source: Source: China NBS

Figure 17: Washing machine production growth



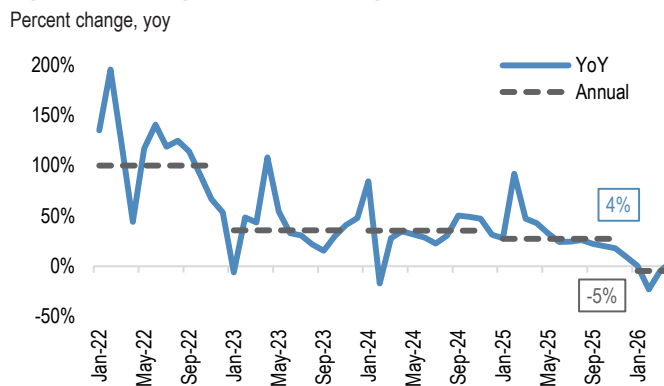
*2026 is through April.
Source: China NBS

Figure 18: Passenger vehicle production growth



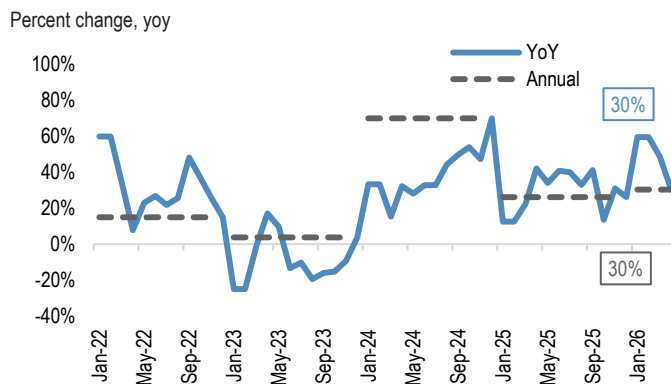
*Includes basic passenger, multi-purpose, SUVs, and crossover vehicles. 2026 is through April.
Source: China Automotive and Research Center

Figure 19: Passenger NEV production growth



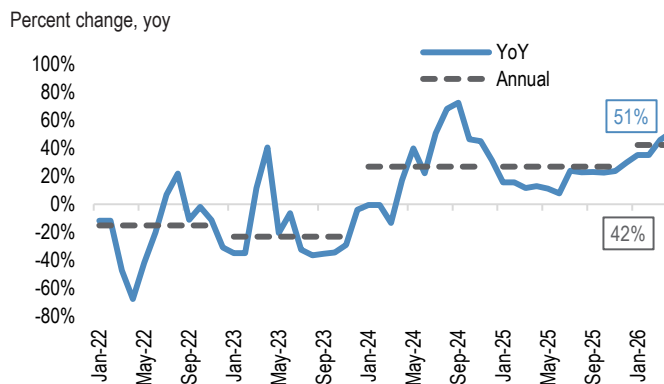
*Includes battery electric vehicles, and plug-in hybrids. 2026 is through April.
Source: China Automotive and Research Center

Figure 20: Industrial robot production growth



*2026 is through April.
Source: China NBS

Figure 21: Excavator production growth

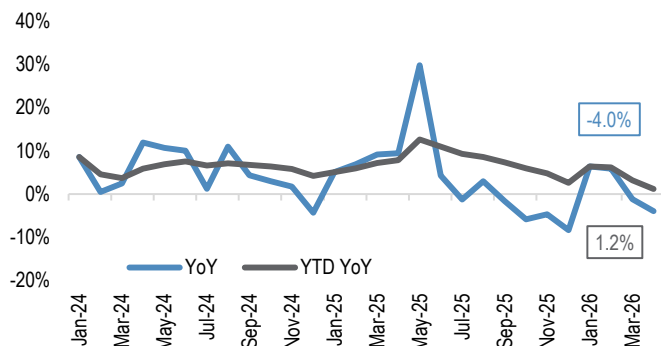


*2026 is through April.
Source: China NBS

Copper

Figure 22: Consumption-weighted end use indicator for Chinese copper demand growth

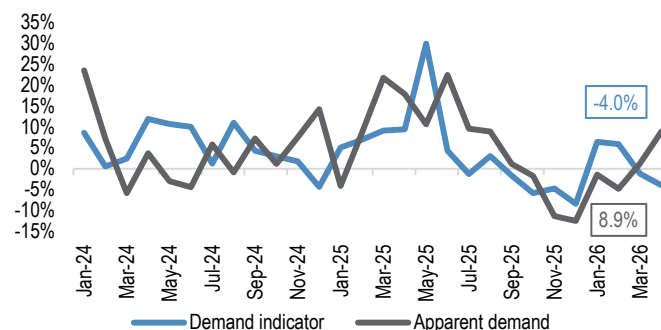
LHS: Percent change, yoy; RHS: Percent change, YTD yoy



*This indicator applies consumption-weights to underlying reported end-use production/installation/export growth figures to derive an estimate for monthly aggregate Chinese copper demand growth. It covers construction, grid, solar, wind, EVs, other transport, consumer durables, ACs/refrigeration, machinery and product and cable exports which, combined, roughly account for 90% of Chinese end use demand. Data through April 2026.
Source: China National Energy Administration, China Automotive Tech and Research Center, Tianjin, NBS, China Customs, Bloomberg Finance L.P., J.P. Morgan Commodities Research

Figure 24: YoY Growth – Consumption-weighted end use copper demand indicator vs apparent demand

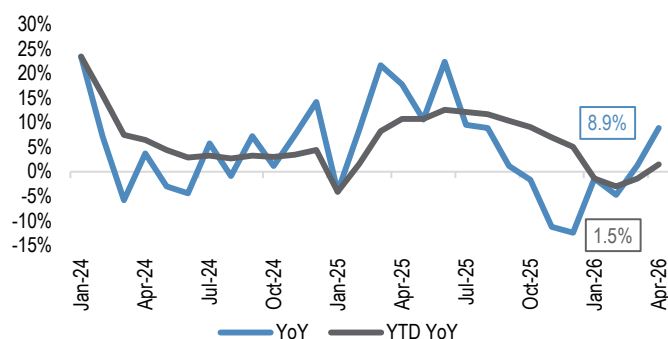
Percent change, yoy



*See explanatory notes in Figures 24 & 25. Data through April 2026.
Source: China National Energy Administration, China Automotive Tech and Research Center, Tianjin, NBS, China Customs, Bloomberg Finance L.P., J.P. Morgan Commodities Research

Figure 23: China copper apparent demand growth

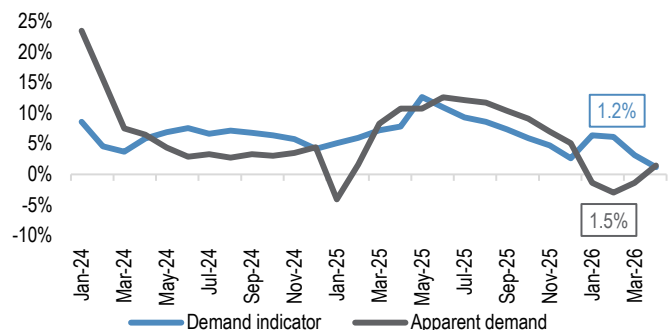
Percent change



*Apparent demand is calculated as (Imports-Exports)+Production-Monthly stock changes on the SHFE. Data through April 2026.
Source: China National Energy Administration, China Automotive Tech and Research Center, Tianjin, NBS, China Customs, Bloomberg Finance L.P., J.P. Morgan Commodities Research

Figure 25: YTD YoY Growth – Consumption-weighted end use copper demand indicator vs apparent demand

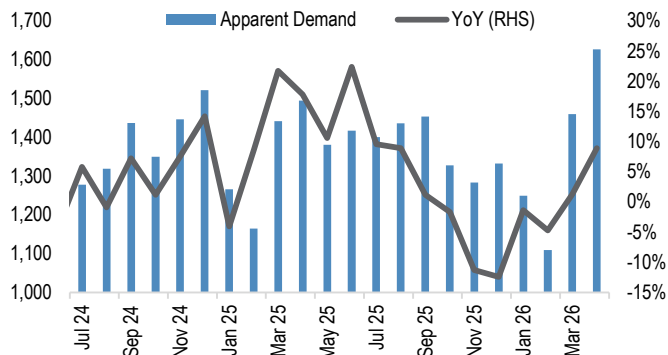
Percent change, YTD yoy



*See explanatory notes in Figures 24 & 25. Data through April 2026.
Source: China National Energy Administration, China Automotive Tech and Research Center, Tianjin, NBS, China Customs, Bloomberg Finance L.P., J.P. Morgan Commodities Research

Figure 26: China apparent copper demand

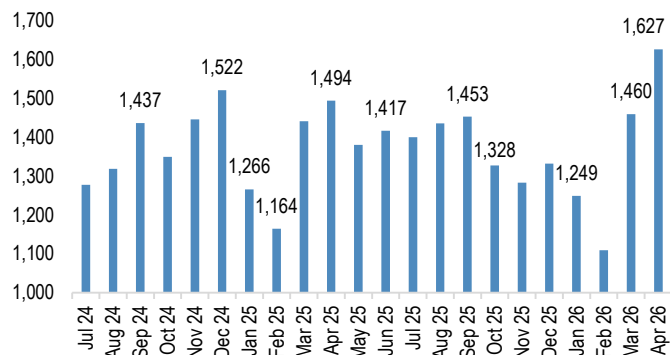
LHS: Thousand mt; RHS: Percent change, yoy



Source: China Customs General Administration, SHFE. *Apparent demand is calculated as (Imports-Exports)+Production-Monthly stock changes on the SHFE.

Figure 27: China apparent copper demand

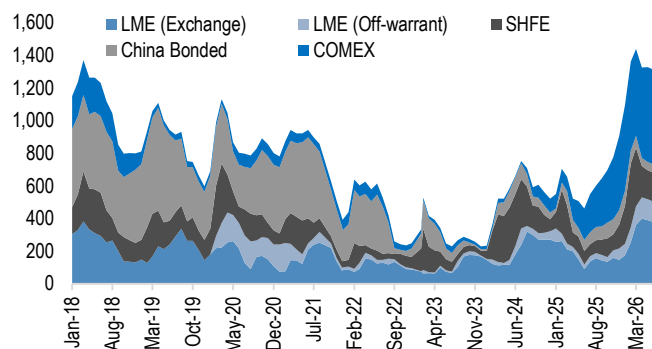
Thousand mt



Source: China Customs General Administration, SHFE. *Apparent demand is calculated as (Imports-Exports)+Production-Monthly stock changes on the SHFE.

Figure 28: Global visible copper inventories

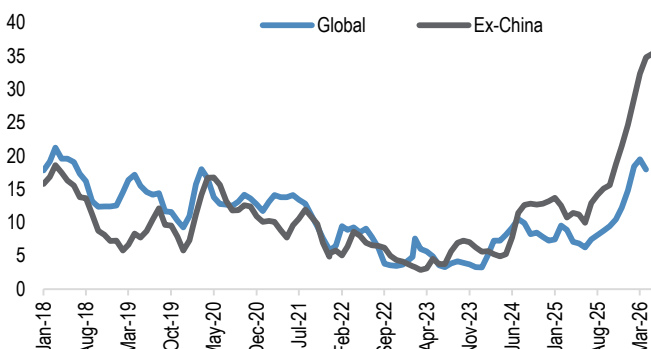
Thousand mt



Source: CRU, LME, SHFE, COMEX. Bonded includes Chinese bonded stocks in Shanghai, Guangdong and Qingdao. As of 4th June 2026, while LME off-warrant as of 1st June 2026.

Figure 29: Global visible copper inventory in days of use

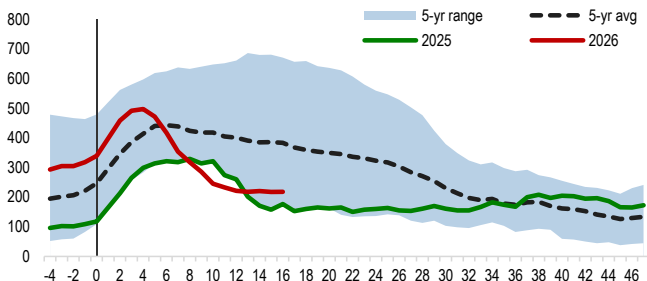
Days



Source: SHFE, CRU, SMM, J.P. Morgan Commodities Research

Figure 30: Total visible China copper inventory (SHFE + Bonded)

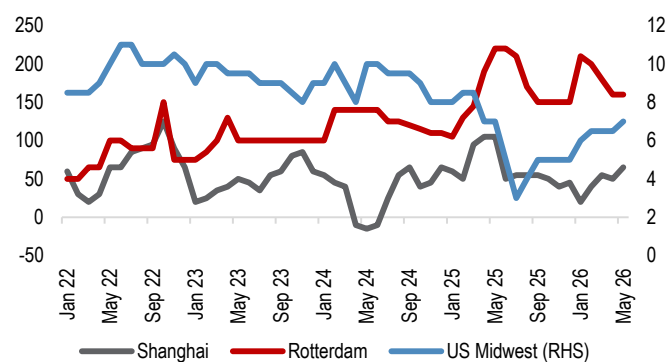
Y-axis: Thousand mt; X-axis: Weeks around Chinese New Year (0 = week closest to start of CNY)



Source: SHFE, CRU, SMM, J.P. Morgan Commodities Research

Figure 31: Copper regional premiums

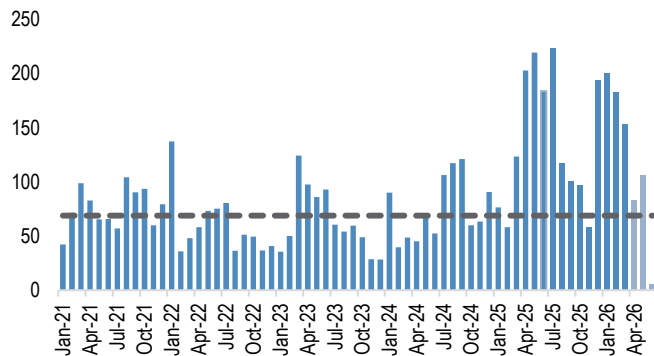
LHS: US\$/t; RHS: US\$/lb



Source: CRU

Figure 32: Monthly US refined copper imports

LHS: Thousand mt, RHS: US\$/mt

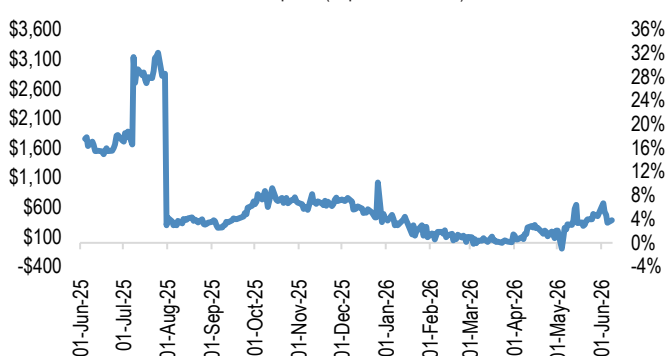


Note: 1st April- 6th June 2026 estimated based on bill of lading data.

Source: US Census Bureau, IHS Markit, COMEX, LME, J.P. Morgan Commodities Research

Figure 33: Jul'26 COMEX/LME copper spread

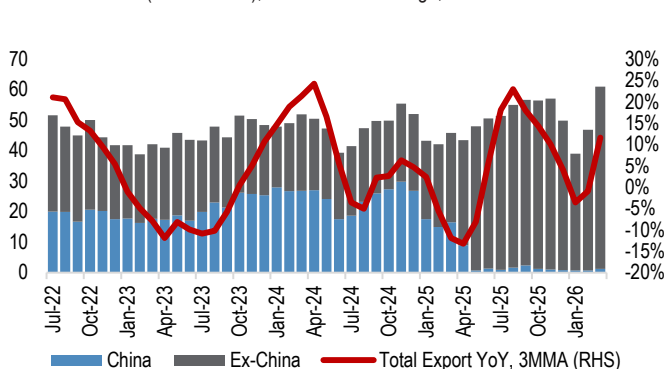
LHS: US\$/mt, RHS: Percent of LME price (implied tariff rate)



Source: Bloomberg Finance L.P. *Arb as of London, 8th June 2026.

Figure 34: Monthly US copper scrap exports

LHS: Thousand mt (cu contained); RHS: Percent change, YoY

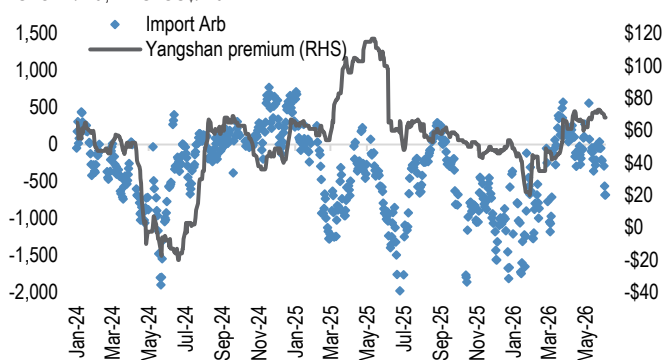


*Data as of March 2026.

Source: US Census Bureau, J.P. Morgan Commodities Research

Figure 35: Chinese spot copper import arbitrage and onshore spot premium/discount

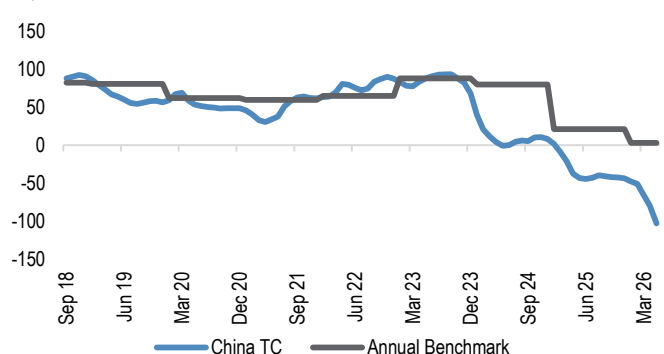
LHS: CNY/mt; RHS: US\$/mt



Source: SMM, Bloomberg Finance L.P., J.P. Morgan Commodities Research. *Arb as of 8:00 AM London. As of 4th June 2026.

Figure 36: China copper imported concentrate treatment charges vs annual benchmark

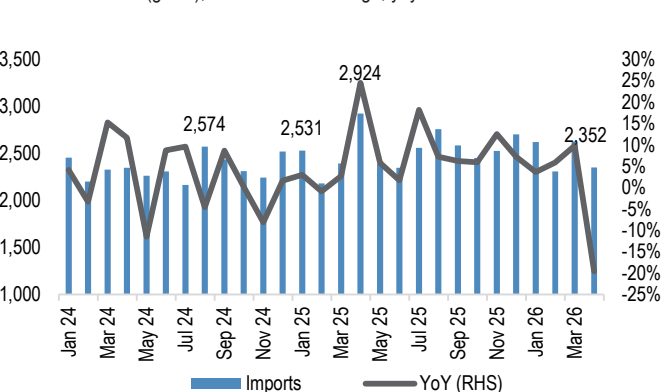
US\$/mt



Source: Metal Bulletin, Platts, Wood Mackenzie

Figure 37: China copper concentrate imports

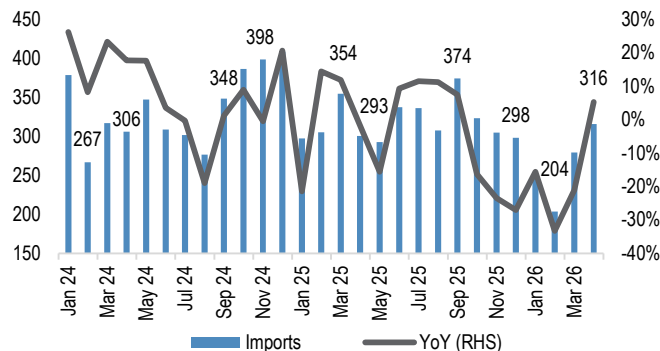
LHS: Thousand mt (gross); RHS: Percent change, yoy



Source: China Customs General Administration

Figure 38: China refined copper imports

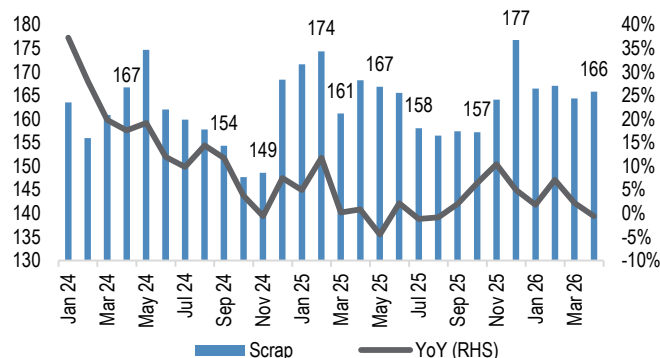
LHS: Thousand mt; RHS: Percent change, yoy



Source: China Customs General Administration

Figure 40: China contained copper scrap imports, 3mma

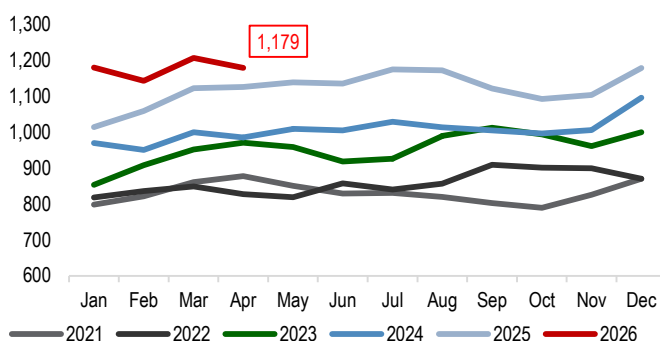
LHS: Thousand mt (Cu contained); RHS: Percent change, yoy



Source: China Customs General Administration

Figure 42: China refined copper production, monthly

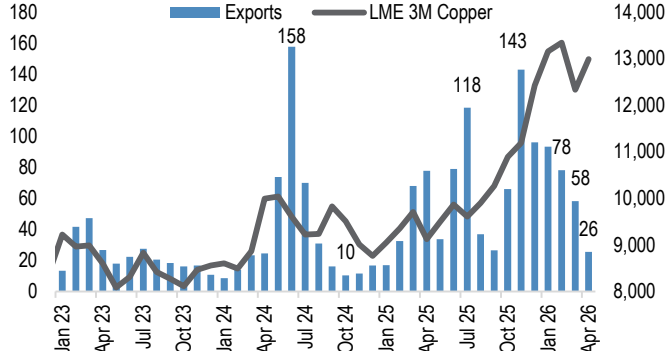
Thousand mt



Source: SMM

Figure 39: China refined copper exports vs LME 3-month copper price

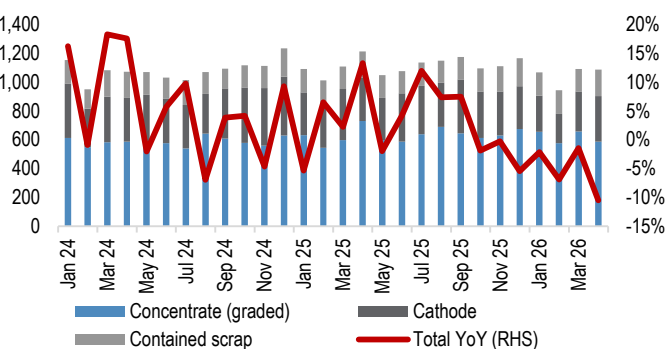
LHS: Thousand mt; RHS: US\$/mt



Source: China Customs General Administration, Bloomberg Finance L.P.

Figure 41: China copper imports of concentrate, cathode and contained scrap

LHS: Thousand mt; RHS: Percent change, yoy

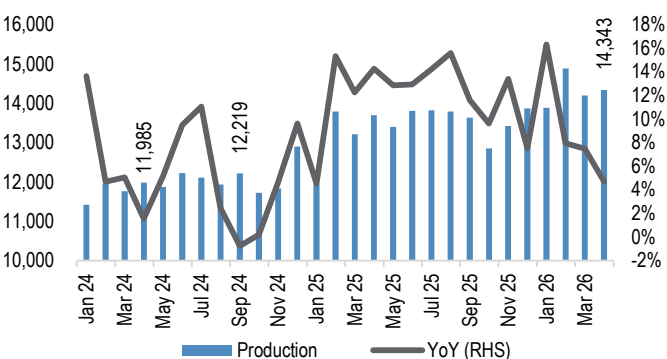


*Assumes 25% copper contained in concentrate.

Source: China Customs General Administration

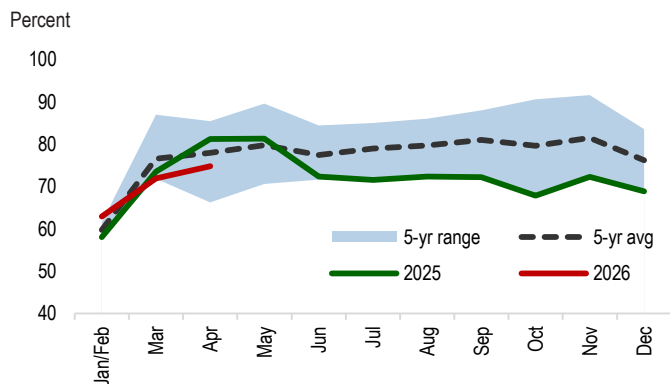
Figure 43: Annualized China refined copper production

LHS: Thousand mt; RHS: Percent change, yoy



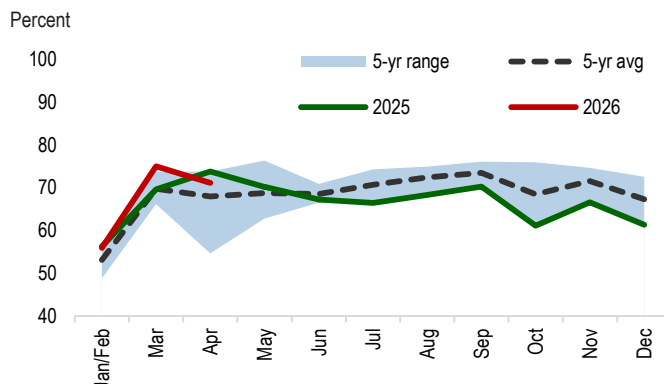
Source: SMM

Figure 44: China copper wire and cable producer operating rates



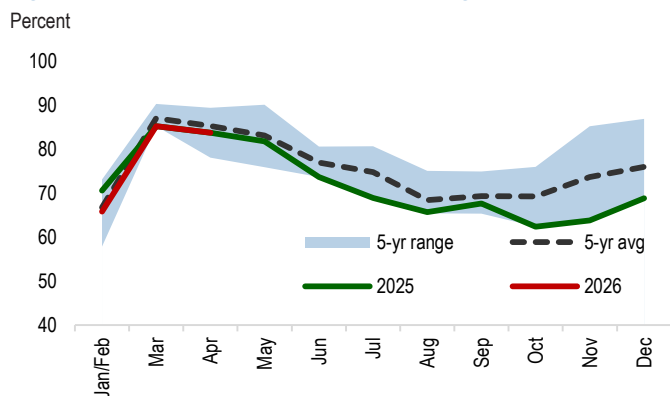
Source: SMM

Figure 45: China copper cathode-fed rod producer operating rates



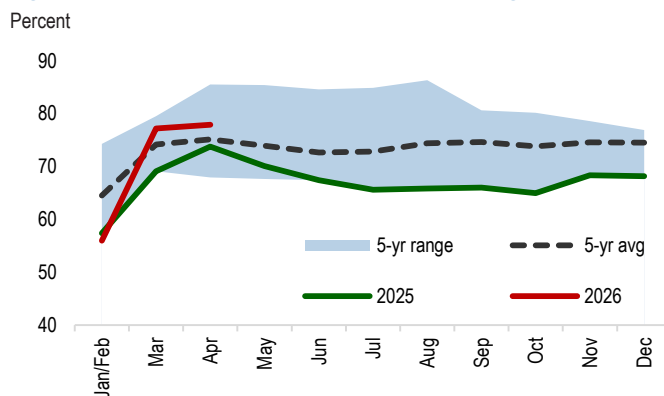
Source: SMM

Figure 46: China copper tube producer operating rates



Source: SMM

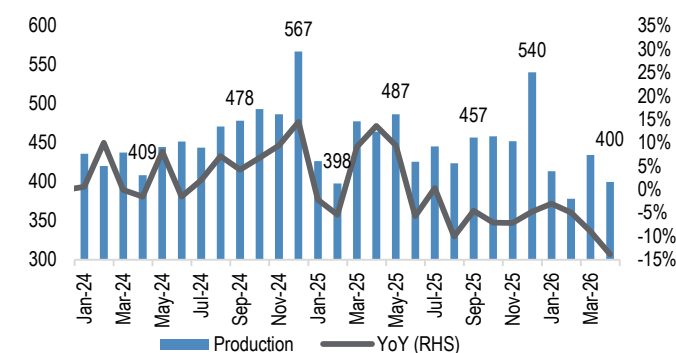
Figure 47: China copper flat-rolled producer operating rates



Source: SMM

Figure 48: Chile copper mine production

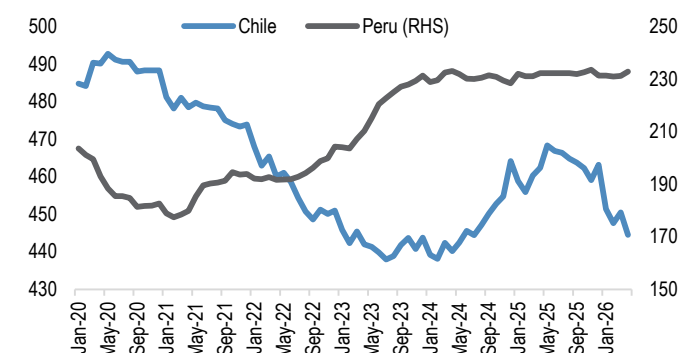
LHS: Thousand metric tonnes; RHS: Percent change yoy



Source: Instituto Nacional de Estadístico

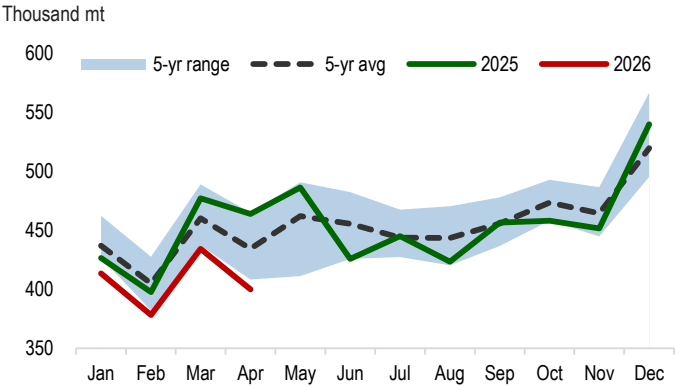
Figure 49: Chile and Peru copper mine production, 12mma

Thousand metric tonnes



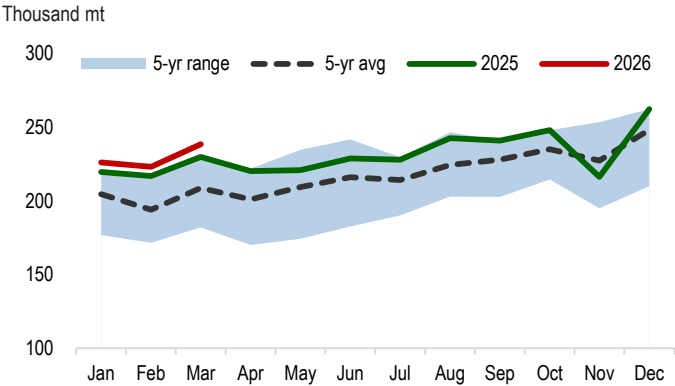
Source: Instituto Nacional de Estadístico, Ministerio de Energías y Minas

Figure 50: Chile copper mine production, monthly



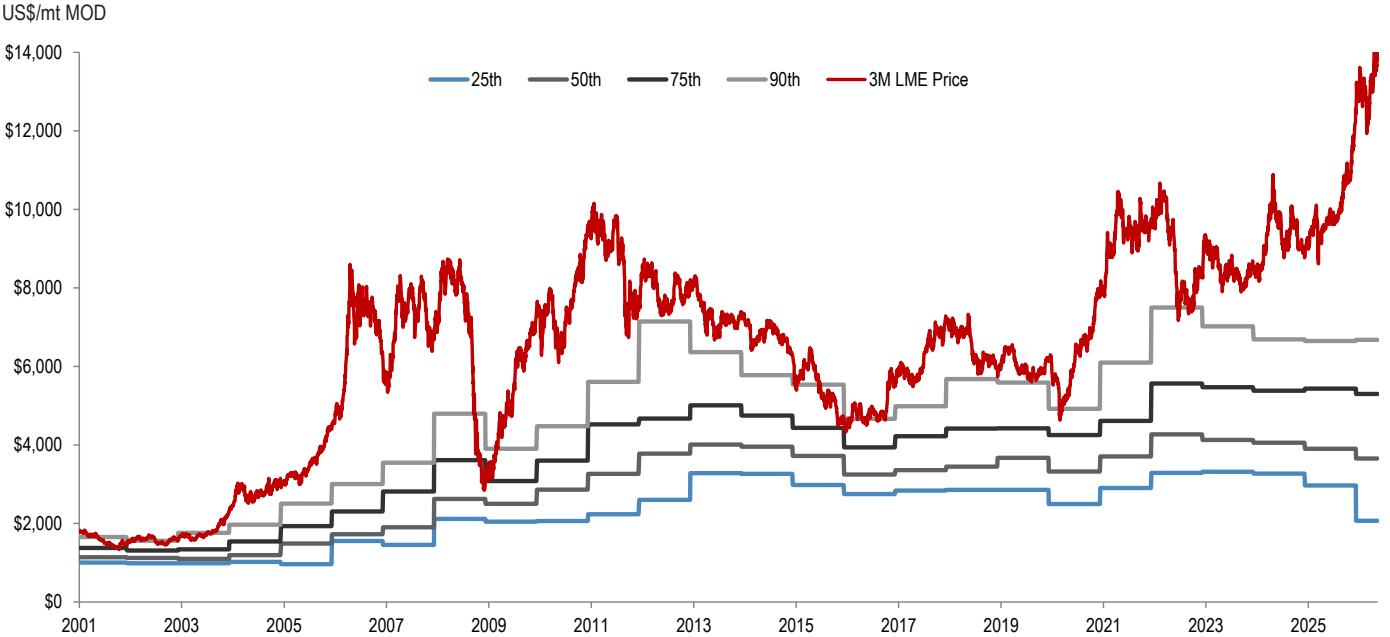
Source: Instituto Nacional de Estadístico

Figure 51: Peru copper mine production, monthly



Source: Ministerio de Energías y Minas

Figure 52: Global copper mining C1 + sustaining capex costs by %-tile and LME 3M copper price



Source: Wood Mackenzie, J.P. Morgan Commodities Research. *LME prices as of 4th June 2026.

Table 5: China copper trade flow

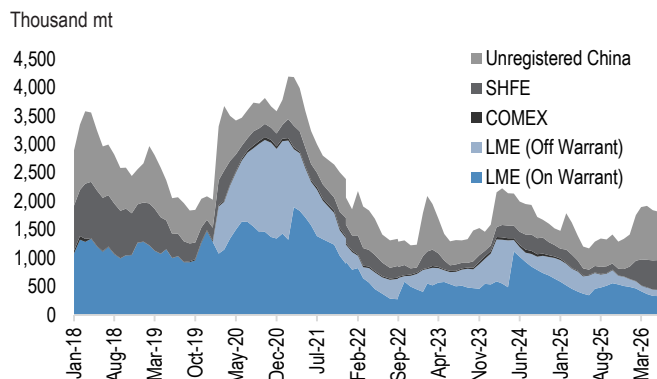
Thousand mt; Percent Change, yoy.

	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD
Copper Concentrate Imports (Gross)	19,720	22,024	21,777	23,428	25,318	27,588	28,165	30,365	9,915
Refined Copper Imports	3,753	3,550	4,670	3,627	3,885	3,739	4,042	3,830	1,049
Unwrought Copper and Products Imports	5,293	4,991	6,686	5,530	5,872	5,504	5,700	5,350	1,570
Contained Copper Imports (from Scrap)	1,441	1,243	733	1,360	1,588	1,721	1,937	1,983	668
Refined Copper Exports	282	316	190	266	253	279	458	795	255
Unwrought Copper & Copper Products Exports	793	841	676	933	917	958	1,287	1,673	608
Net Refined Imports	3,472	3,234	4,480	3,361	3,632	3,459	3,584	3,035	794
YoY-%									
Copper Concentrate Imports		11.7%	-1.1%	7.6%	8.1%	9.0%	2.1%	7.8%	-1.2%
Refined Copper Imports		-5.4%	31.5%	-22.3%	7.1%	-3.8%	8.1%	-5.3%	-16.5%
Unwrought Copper and Products Imports		-5.7%	34.0%	-17.3%	6.2%	-6.3%	3.6%	-6.1%	-10.3%
Contained Copper Imports (from scrap)		-13.7%	-41.0%	85.4%	16.8%	8.4%	12.5%	2.4%	0.0%
Refined Copper Exports		12.4%	-40.1%	40.5%	-5.2%	10.6%	63.9%	73.6%	30.8%
Unwrought Copper & Copper Products Exports		6.1%	-19.6%	38.0%	-1.8%	4.5%	34.3%	30.0%	26.6%
Net Refined Imports		-6.8%	38.5%	-25.0%	8.1%	-4.8%	3.6%	24.9%	-20.9%

*Assumes 25% copper contained in concentrate. 2026 is through April.
Source: China Customs General Administration

Aluminium

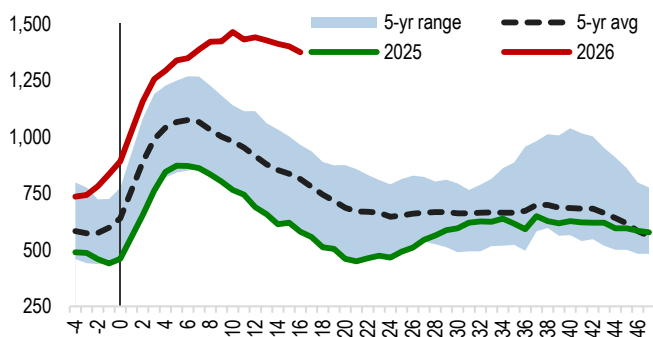
Figure 53: Global visible aluminium inventories



Source: CRU, LME, SHFE, COMEX. Regional includes stocks in Shanghai, Jiangsu, Nanhai, Zhejiang, Tianjin, Henan, Shandong, Chongqing. As of 5th June 2026.

Figure 55: Total visible China aluminium inventory (SHFE + regional warehouses)

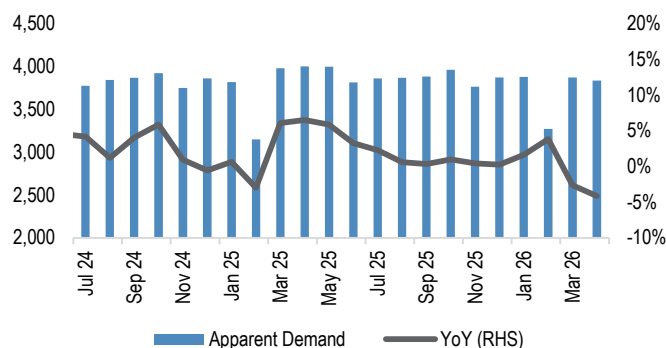
Y-axis: Thousand mt; X-axis: Weeks around Chinese New Year (0 = week closest to start of CNY)



Source: SHFE, CRU, SMM, J.P. Morgan Commodities Research

Figure 57: China apparent aluminium demand

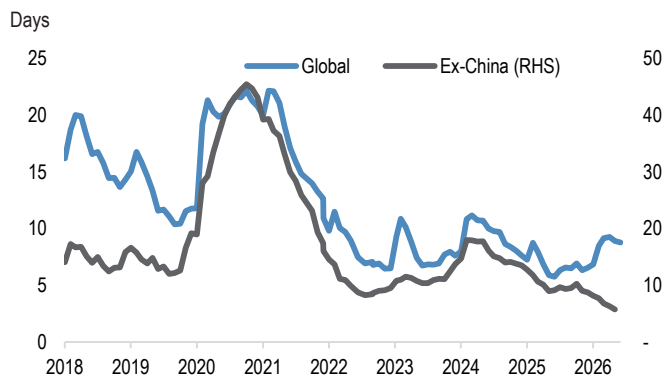
LHS: Thousand mt; RHS: Percent change, yoy



*Apparent demand is calculated as (Imports-Exports)+Production-Monthly stock changes on the SHFE.

Source: CRU, China Customs General Administration, SHFE

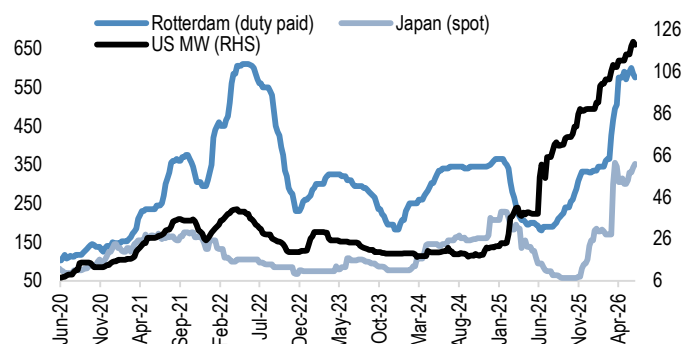
Figure 54: Global visible aluminium inventory in days of use



Source: SHFE, SMM, J.P. Morgan Commodities Research

Figure 56: Regional aluminium ingot premiums

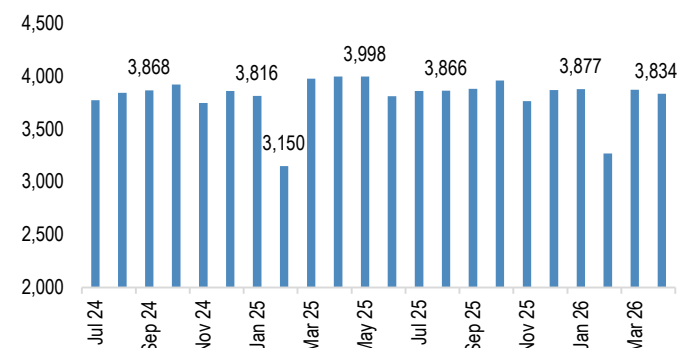
LHS: US\$/mt; RHS: c/lb



Source: CRU

Figure 58: China apparent aluminium demand

Thousand metric tonnes

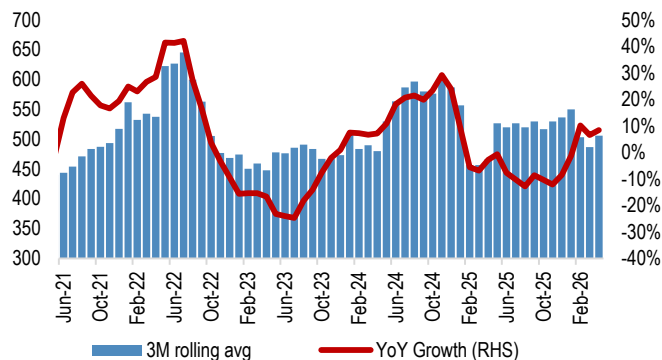


*Apparent demand is calculated as (Imports-Exports)+Production-Monthly stock changes on the SHFE.

Source: CRU, China Customs General Administration, SHFE

Figure 59: Chinese unwrought aluminium and aluminium product exports, 3-month rolling average

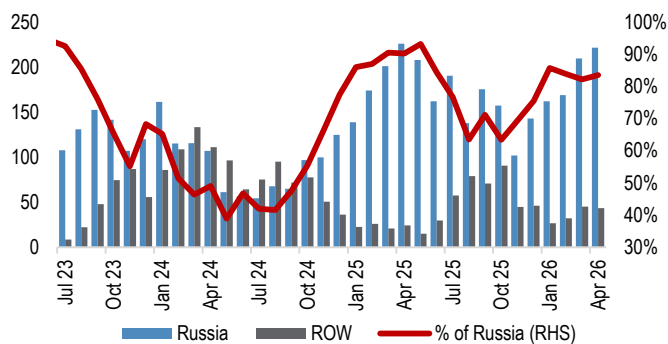
LHS: Thousand mt; RHS: Percent change, yoy



Source: China Customs, J.P. Morgan Commodities Research

Figure 61: China refined aluminium imports by region

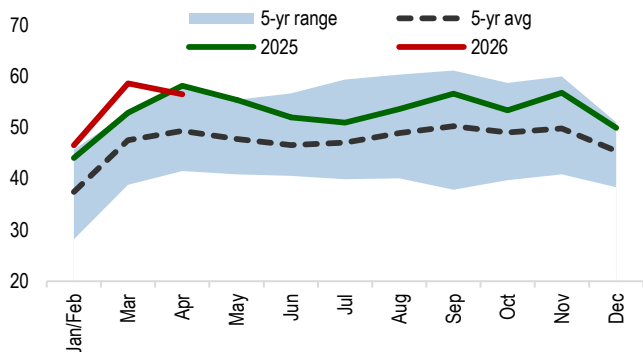
LHS: Thousand mt; RHS: Percent of total



Source: China Customs, J.P. Morgan Commodities Research

Figure 63: China aluminium wire & rod producer operating rates

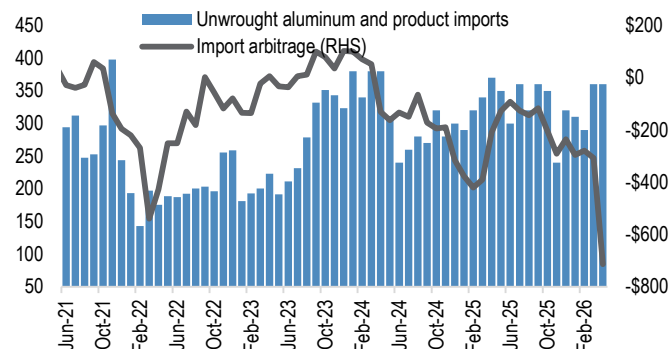
Percent



Source: SMM, Bloomberg Finance L.P.

Figure 60: Chinese unwrought aluminium and aluminium product imports

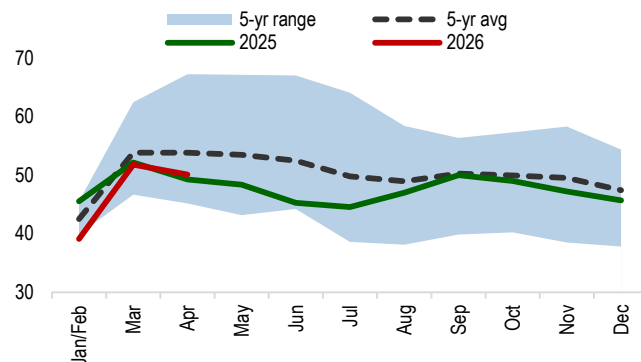
LHS: Thousand mt; RHS: US\$/mt



Source: China Customs, J.P. Morgan Commodities Research

Figure 62: China aluminium profile producer operating rates

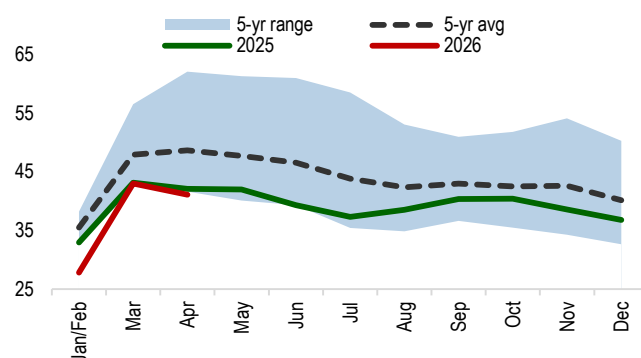
Percent



Source: SMM, Bloomberg Finance L.P.

Figure 64: China aluminium extrusions producer operating rates

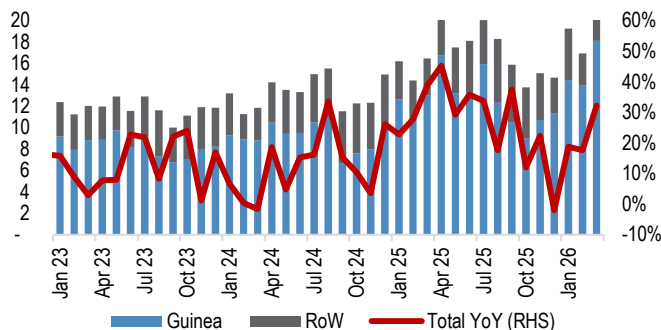
Percent



Source: SMM, Bloomberg Finance L.P.

Figure 65: China total bauxite imports

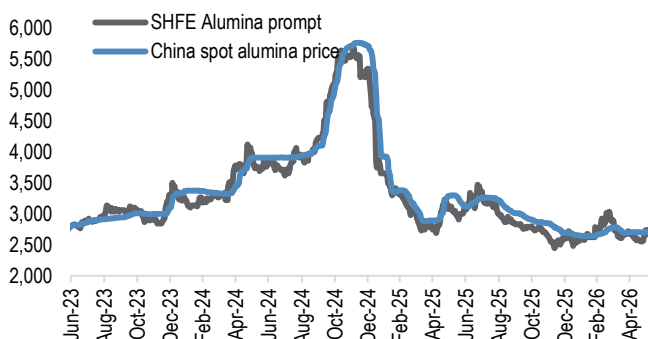
LHS: Million mt; RHS: Percent change yoy



Source: China Customs

Figure 67: China alumina price

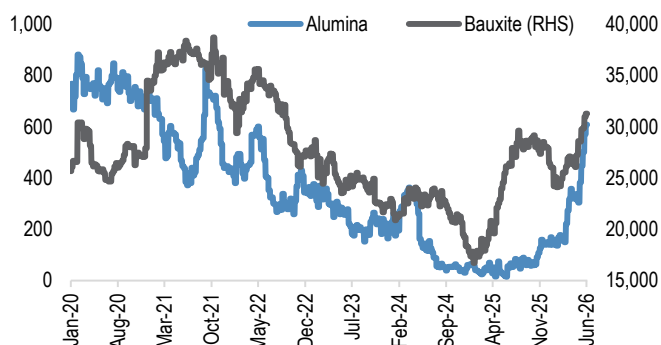
CNY/mt



Source: SHFE, Asian Metal Inc, Bloomberg Finance L.P.

Figure 69: China port inventory of bauxite and alumina

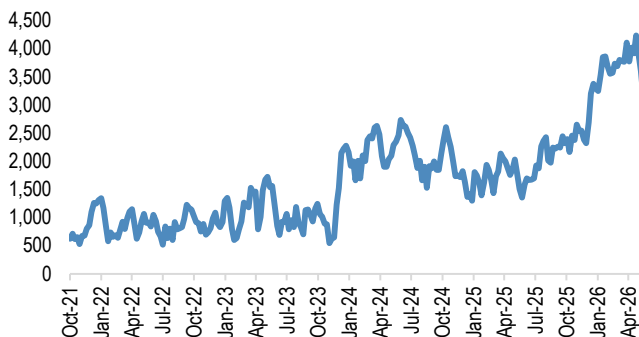
Thousand mt



Source: Mysteel Global

Figure 66: China imports of bauxite from Guinea, 3-week ma

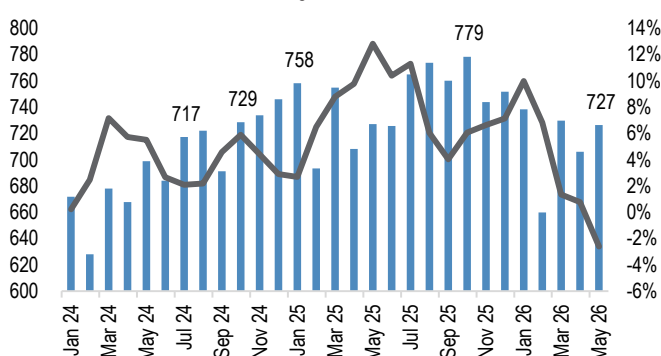
Thousand mt



Source: Mysteel Global

Figure 68: China monthly alumina production

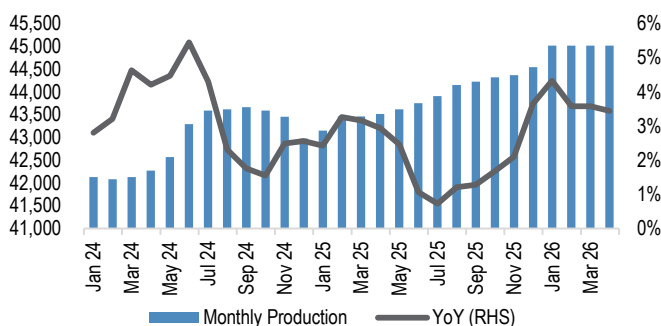
LHS: Thousand mt; RHS: Percent change



Source: SMM, Bloomberg Finance L.P.

Figure 70: Annualized Chinese refined monthly aluminium production

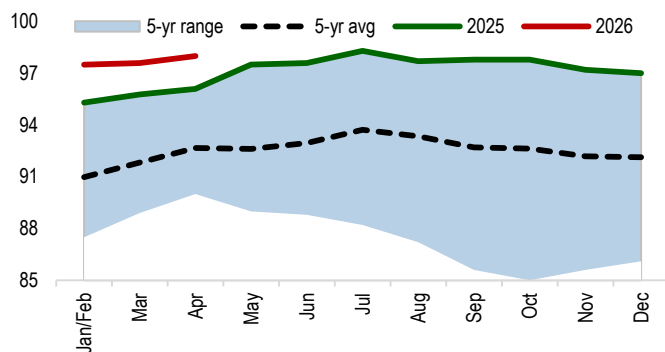
LHS: Thousand mt; RHS: Percent change



Source: CRU

Figure 71: China aluminium ingot producer operating rates

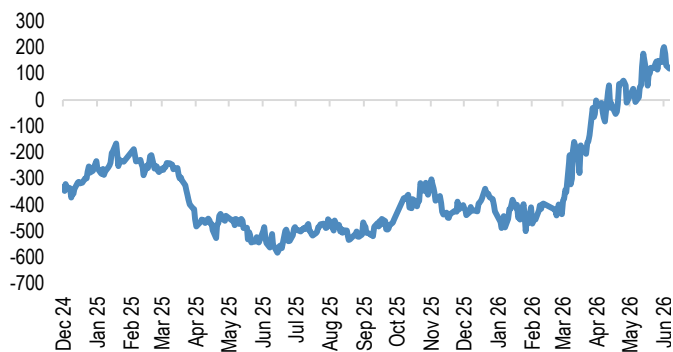
Percent



Source: SMM, Bloomberg Finance L.P.

Figure 72: China aluminium semis export arb proxy (China to rest of Asia)

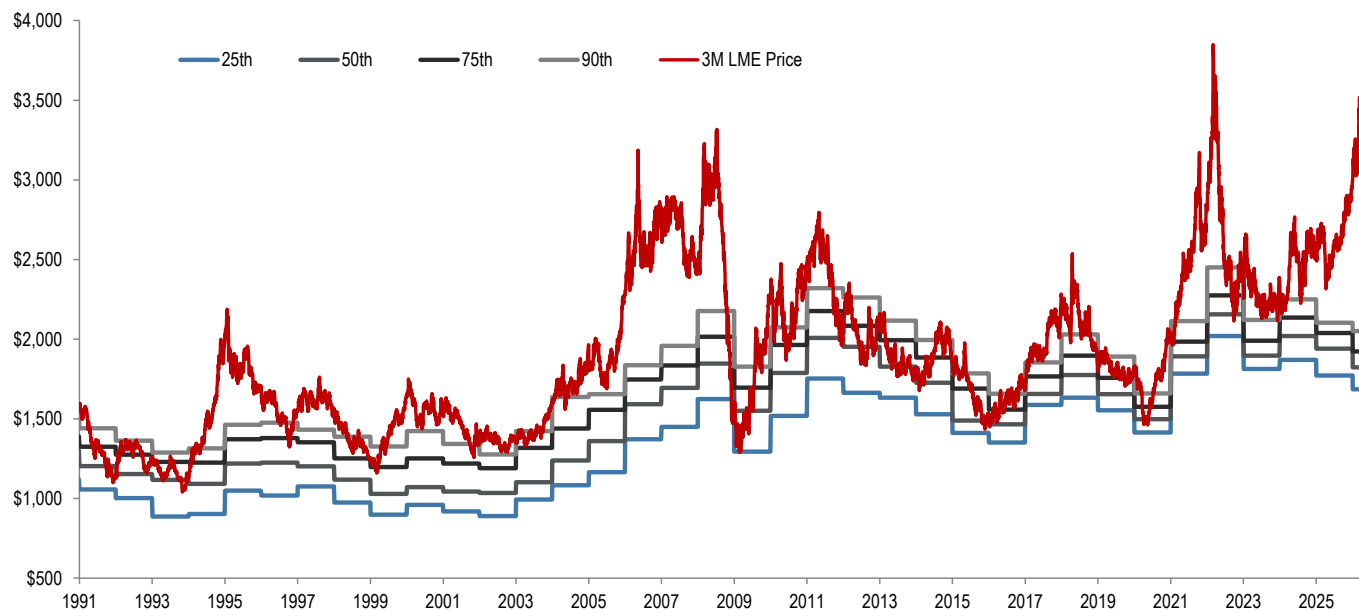
US\$/mt. Differential between China spot prices (inclusive of VAT) and LME + Asia premium



Note: Export arb is calculated as the differential between China spot prices and LME + Asia premium less conversion costs (ingot to semis), logistics, shipping and financing costs.
Source: SHFE, SMM, Shanghai Metals Exchange, LME, Bloomberg L.P., J.P. Morgan Commodities Research. *Prices as of 4th June 2026.

Figure 73: Global primary aluminium smelting C1 costs by %-tile and LME 3M aluminium price

US\$/mt MOD



Source: Wood Mackenzie, LME, J.P. Morgan Commodities Research. *LME prices as of 4th June 2026.

Table 6: China aluminium trade flow

Thousand mt; Percent Change, yoy.

	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD
Bauxite Imports	82,689	100,665	111,537	107,421	125,689	141,647	159,179	201,185	77,728
Refined Aluminium Imports	125	75	1,064	1,580	668	1,543	2,136	2,547	911
Unwrought Aluminium and Products Imports	596	646	2,704	3,214	2,391	3,060	3,740	3,920	1,320
Aluminum Waste and Scrap Imports	1,565	1,395	825	1,033	1,518	1,753	1,785	2,016	171
Refined Aluminium Exports	55	76	10	7	196	150	121	298	16
Unwrought Aluminium & Aluminium Products Exports	5,800	5,735	4,860	5,620	6,604	5,679	6,660	6,130	598
Net Refined Imports	70	-1	1,055	1,573	472	1,393	2,015	2,249	896
YoY-%									
Bauxite Imports		21.7%	10.8%	-3.7%	17.0%	12.7%	12.4%	26.4%	14.7%
Refined Aluminium Imports		-39.5%	1309.8%	48.4%	-57.7%	131.0%	38.4%	19.2%	9.2%
Unwrought Aluminium and Products Imports		8.3%	318.8%	18.8%	-25.6%	28.0%	22.2%	4.8%	0.0%
Aluminum Waste and Scrap Imports		-10.8%	-40.9%	25.3%	46.9%	15.5%	1.8%	12.9%	-75.5%
Refined Aluminium Exports		38.0%	-87.3%	-27.1%	2680.0%	-23.4%	-19.4%	145.7%	-75.5%
Unwrought Aluminium & Aluminium Products Exports		-1.1%	-15.3%	15.6%	17.5%	-14.0%	17.3%	-8.0%	-68.4%

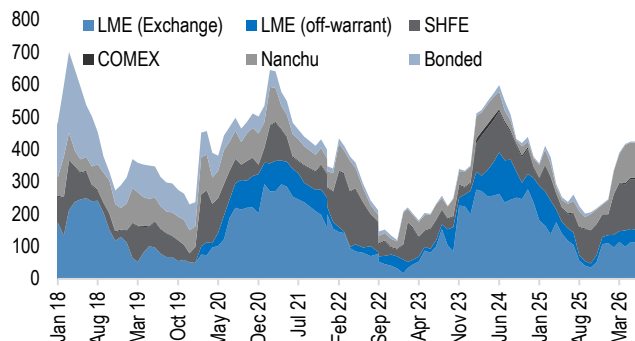
*2026 is through April.

Source: China Customs General Administration

Zinc

Figure 74: Global visible zinc inventories

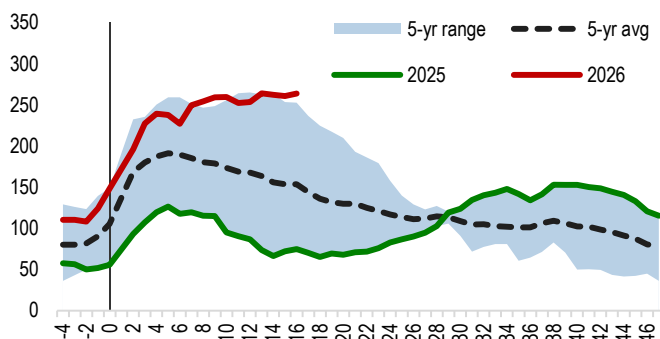
Thousand metric tonnes



Source: CRU, LME, SHFE, COMEX. Nanchu and Bonded warehouses are in China. As of 4th June 2026.

Figure 76: Total visible China zinc inventory (SHFE + Regional Warehouses)

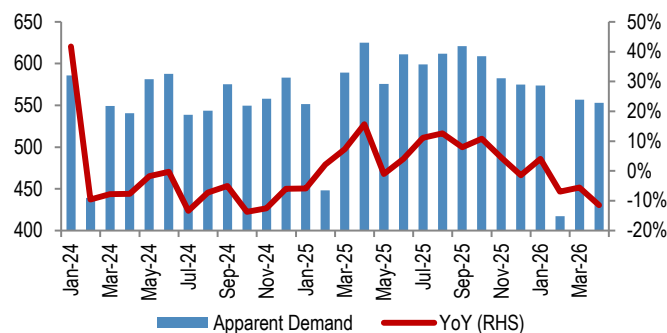
Y-axis: Thousand mt; X-axis: Weeks around Chinese New Year (0 = week closest to start of CNY)



Source: SHFE, SMM, J.P. Morgan Commodities Research

Figure 78: China apparent zinc demand

LHS: Thousand mt; RHS: Percent change, yoy

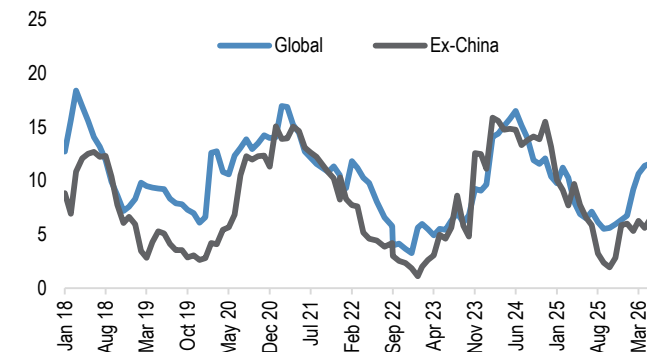


*Apparent demand is calculated as (Imports-Exports)+Production-Monthly stock changes on the SHFE

Source: China Customs General Administration, SHFE

Figure 75: Global visible zinc inventories in days of use

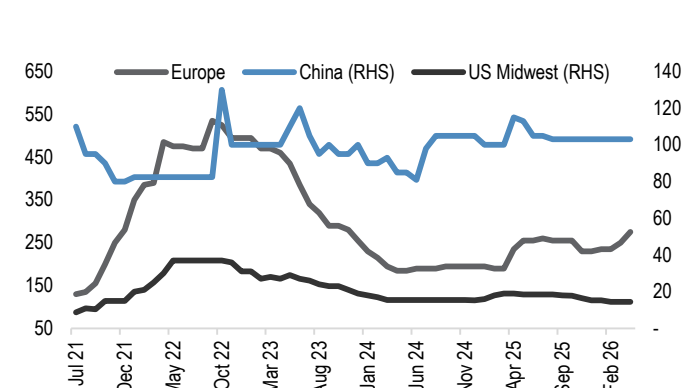
Days



Source: SHFE, SMM, J.P. Morgan Commodities Research

Figure 77: Regional zinc premiums

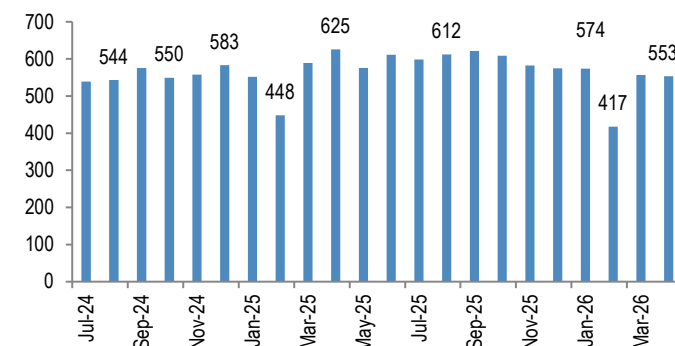
US\$/mt



Source: CRU

Figure 79: China apparent zinc demand

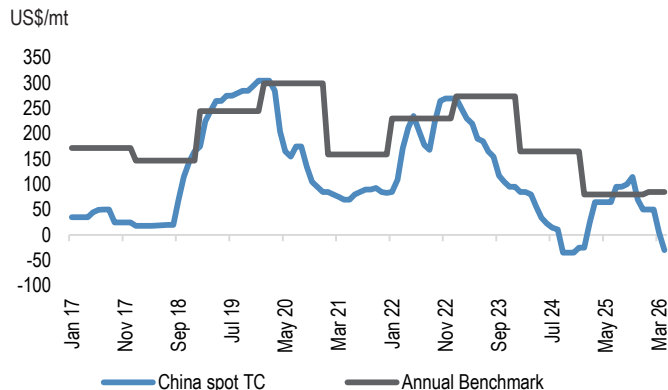
Thousand mt



*Apparent demand is calculated as (Imports-Exports)+Production-Monthly stock changes on the SHFE.

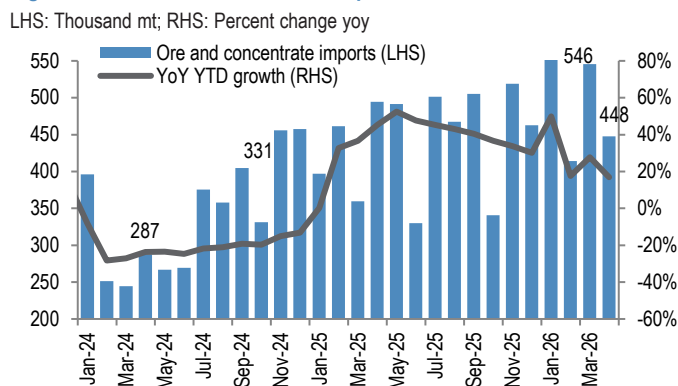
Source: China Customs General Administration, SHFE

Figure 80: China zinc spot concentrate treatment charges vs annual benchmark



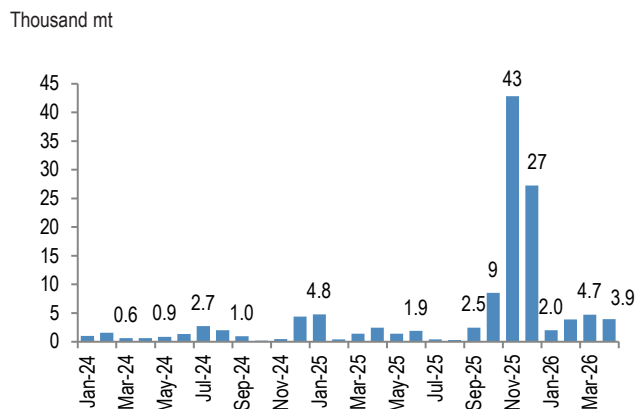
Source: Metal Bulletin, Platts, Wood Mackenzie

Figure 82: China zinc concentrate imports



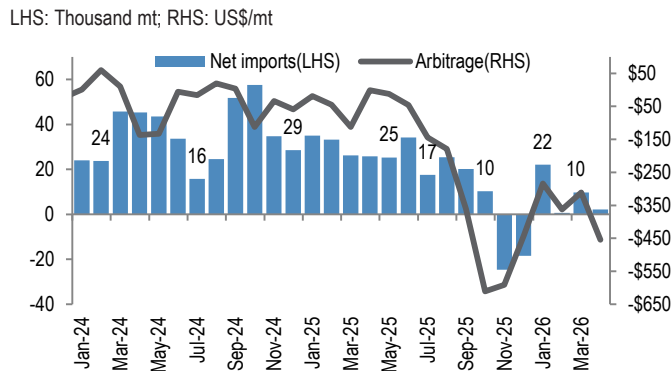
Source: NBS, Bloomberg Finance L.P.

Figure 84: China refined zinc exports



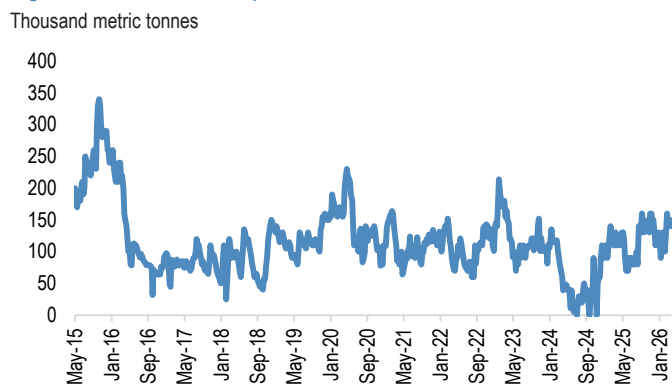
Source: NBS, Bloomberg Finance L.P.

Figure 81: China net imports of refined zinc and China's zinc import arb



Source: China Customs, CRU, J.P. Morgan Commodities Research

Figure 83: China zinc ore port inventories

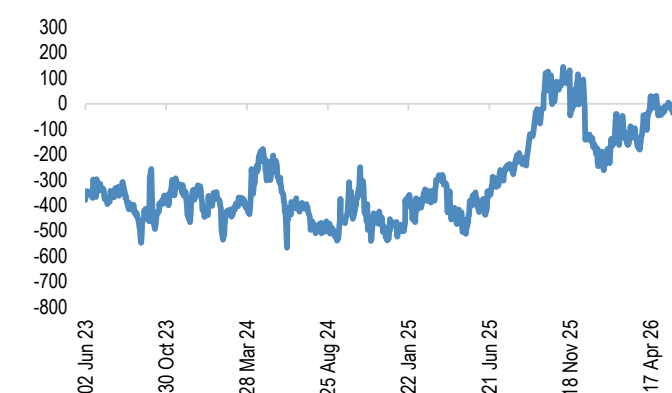


*Concentrate inventory at the Lianyungang port.

Source: SMM

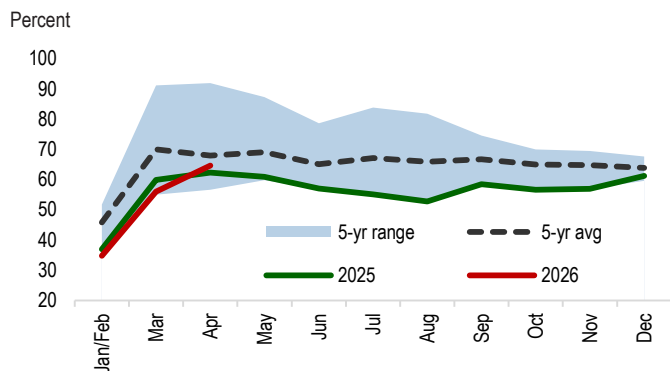
Figure 85: China zinc export arb proxy

US\$/mt. Differential between China spot prices (inclusive of VAT) and LME 3M + Asia premium



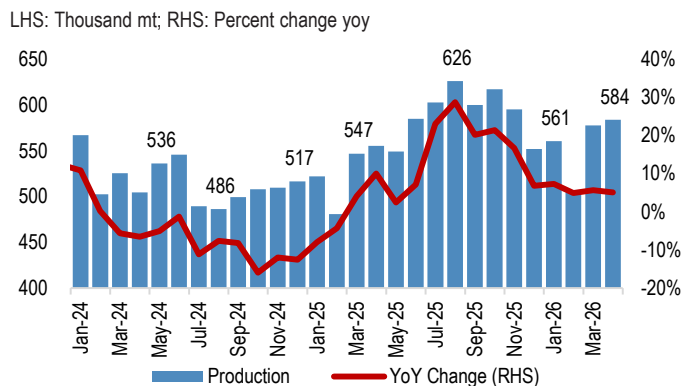
Source: SHFE, SMM, Shanghai Metals Exchange, LME, Bloomberg L.P., J.P. Morgan Commodities Research. *Prices as of 4th June 2026.

Figure 86: China zinc galvanizers operating rates



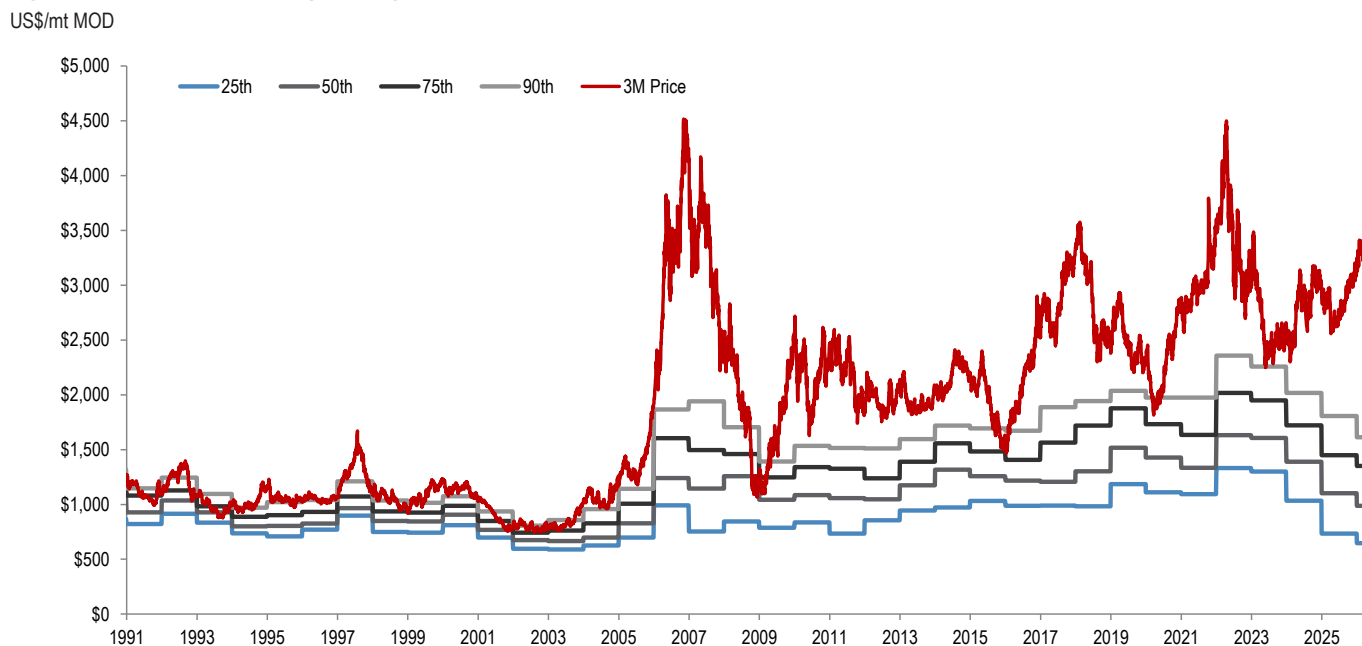
Source: SMM, J.P. Morgan Commodities Research

Figure 87: China monthly refined zinc production



Source: SMM, Bloomberg Finance L.P.

Figure 88: Global zinc C1 mining costs by %-tile and LME 3M copper price



Source: Wood Mackenzie, LME, J.P. Morgan Commodities Research. *LME prices as of 4th June 2026.

Table 7: China zinc trade flow

Thousand mt; Percent Change, yoy.

	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD
Concentrate Imports (Gross)	2,969	3,183	3,822	3,648	4,135	4,723	4,097	5,331	2,002
Refined Zinc Imports	715	605	541	434	79	380	446	304	49
Unwrought Zinc and Alloys Imports	821	692	615	520	142	429	495	348	60
Refined Zinc Exports	22	62	29	5	81	9	17	94	15
Unwrought Zinc and Alloys Exports	24	64	32	7	83	12	22	101	15
Net Refined Imports	693	544	512	429	-2	371	429	210	35
YoY-%									
Concentrate Imports (Gross)		7.2%	20.1%	-4.6%	13.3%	14.2%	-13.3%	30.1%	16.9%
Refined Zinc Imports		-15.4%	-10.6%	-19.8%	-81.8%	380.1%	17.2%	-31.8%	-62.0%
Unwrought Zinc and Alloys Imports		-15.7%	-11.2%	-15.4%	-72.6%	201.7%	15.3%	-29.8%	-58.5%
Refined Zinc Exports		181.3%	-52.5%	-81.8%	1415.5%	-89.0%	88.2%	460.0%	59.7%
Unwrought Zinc and Alloys Exports		163.9%	-50.8%	-79.0%	1154.1%	-85.1%	78.4%	358.4%	28.7%

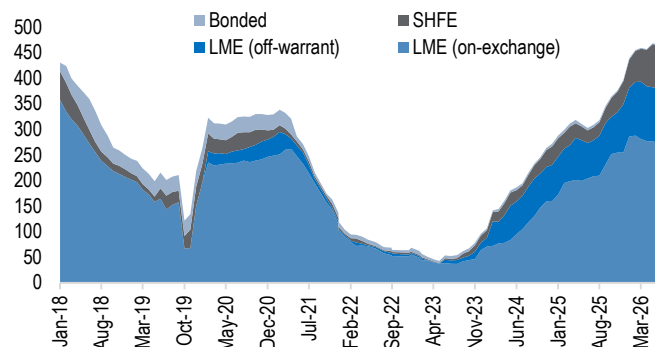
*2026 is through April.

Source: China Customs General Administration

Nickel

Figure 89: Global visible nickel inventory

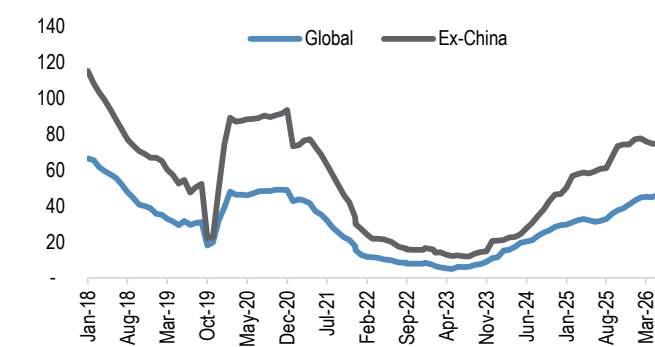
Thousand mt



Source: LME, SHFE, COMEX. Bonded warehouses are in China. As of 4th June 2026.

Figure 90: Global visible nickel inventory in days of use

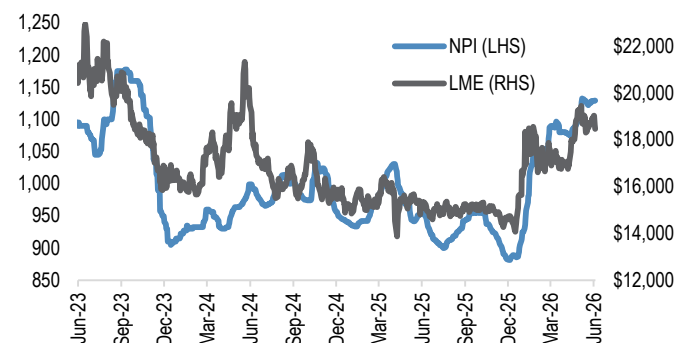
Days



Source: SHFE, J.P. Morgan Commodities Research

Figure 91: Chinese nickel pig iron prices (8-12%) and LME cash nickel prices

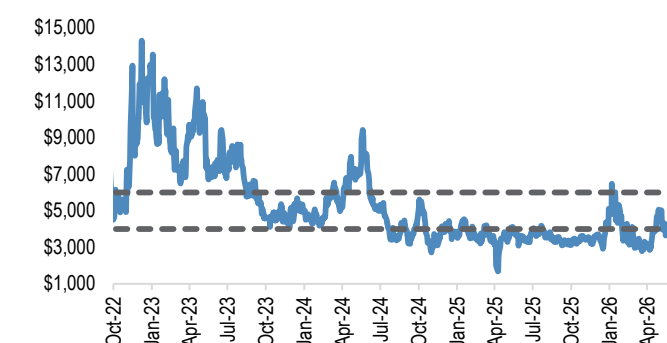
LHS: CNY/nickel tonne unit; RHS: US\$/mt



Source: SMM, LME, Bloomberg Finance L.P., J.P. Morgan Commodities Research. As of 4th June 2026.

Figure 92: Discount spread of NPI prices compared to LME cash nickel prices

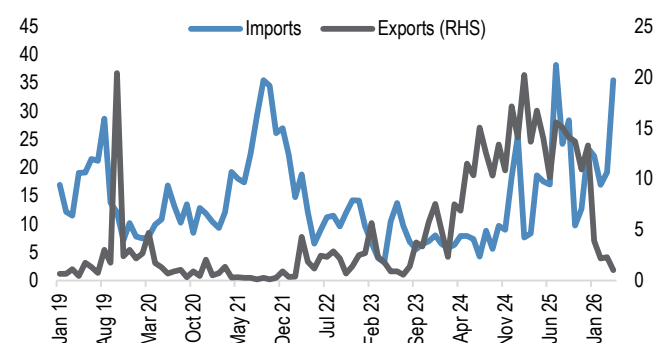
US\$/mt. Dashed lines indicates level of estimated conversion costs



Source: SMM, LME, Bloomberg Finance L.P., J.P. Morgan Commodities Research. As of 4th June 2026.

Figure 93: China monthly refined nickel imports and exports

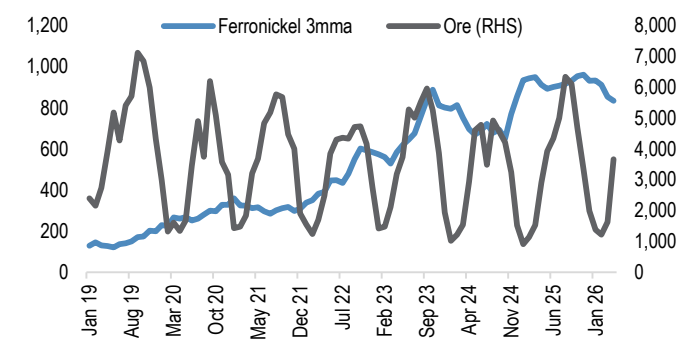
Thousand mt



Source: China Customs, Bloomberg Finance L.P.

Figure 94: China monthly ferronickel and nickel ore imports

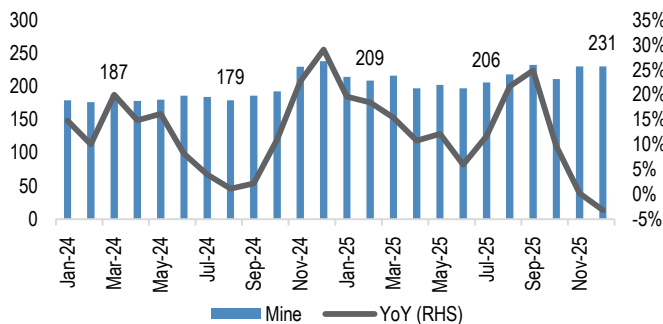
Thousand mt



Source: China Customs, Bloomberg Finance L.P.

Figure 95: Indonesia monthly nickel mine production

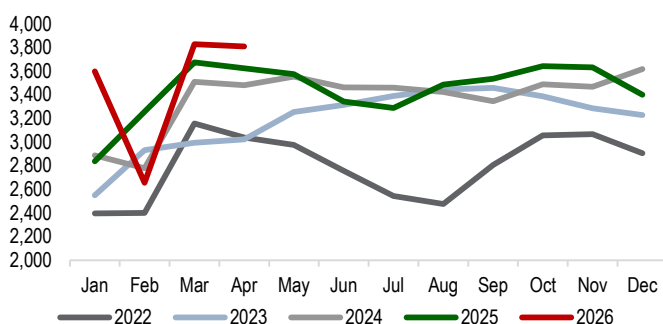
LHS: Thousand mt; RHS: Percent change, yoy



Source: Bloomberg Finance L.P.

Figure 97: China stainless steel production

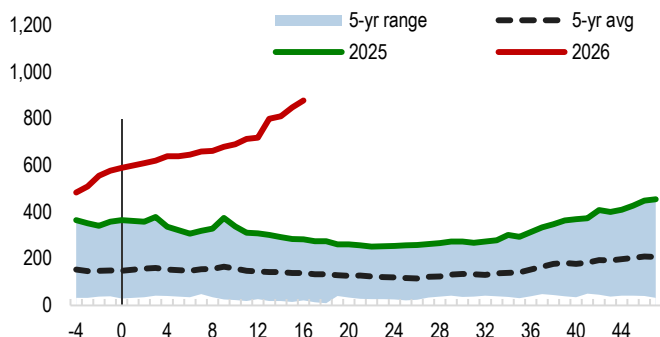
Thousand mt



Source: CRU

Figure 99: Total visible China nickel inventory (SHFE + Regional Warehouses)

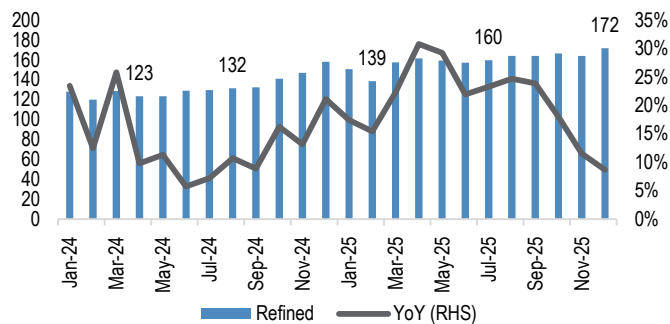
Y-axis: Thousand mt; X-axis: Weeks around Chinese New Year (0 = week closest to start of CNY)



Source: SHFE, SMM, J.P. Morgan Commodities Research

Figure 96: Indonesia monthly refined nickel production

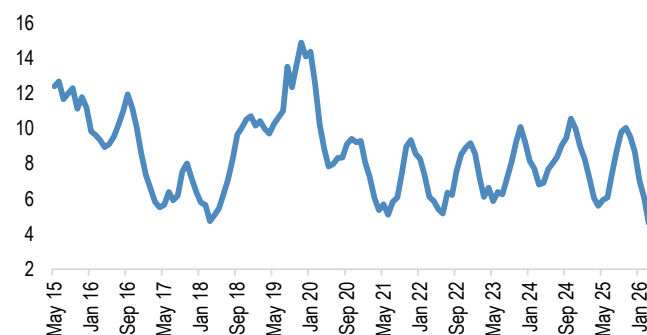
LHS: Thousand mt; RHS: Percent change, yoy



Source: Bloomberg Finance L.P.

Figure 98: China nickel ore port inventory

Million tonnes



Source: China Customs, Bloomberg Finance L.P. *In wet metric tonnes, with nickel content of 0.86%.

Figure 100: Global nickel industry C1 costs by %-tile and LME 3M nickel price

US\$/mt MOD



Source: Wood Mackenzie, LME, J.P. Morgan Commodities Research. *LME prices as of 4th June 2026.

Table 8: China nickel trade flow

Thousand mt; Percent Change, yoy

	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD
Nickel Ore Imports	46,971	56,116	39,104	43,529	40,241	44,819	38,096	42,263	7,904
Refined Nickel Imports	212	193	131	261	156	91	97	231	94
Refined Nickel Exports	13	37	19	5	23	37	123	171	9
Net Refined Imports	199	156	112	256	133	54	-25	60	84
YoY-%									
Nickel Ore Imports		19.5%	-30.3%	11.3%	-7.6%	11.4%	-15.0%	10.9%	21.5%
Refined Nickel Imports		-8.9%	-32.3%	99.8%	-40.3%	-41.6%	6.8%	137.7%	55.8%
Refined Nickel Exports		182.5%	-49.3%	-71.3%	314.2%	62.7%	235.0%	39.7%	-85.5%

*2026 is through April.

Source: China Customs General Administration

JPM metals price forecasts

Quarterly and annual averages

	4Q25A	1Q26A	2Q26	3Q26	4Q26	1Q27	2Q27	3Q27	4Q27	2025	2026	2027
Base Metals												
Aluminum	2,828	3,198	3,800	3,600	3,400	3,200	3,000	2,900	2,750	2,631	3,500	2,963
Copper	11,105	12,824	13,500	13,000	12,500	11,800	11,600	11,600	11,500	9,947	12,956	11,625
Nickel	14,892	17,338	17,500	17,000	17,250	17,500	18,000	18,000	17,500	15,164	17,272	17,750
Zinc	3,165	3,237	3,420	3,500	3,400	3,300	3,200	3,050	3,000	2,867	3,389	3,138
Precious Metals												
Gold	4,152	4,873	4,800	5,300	6,000	6,200	6,250	6,300	6,300	3,440	5,243	6,263
Silver	55.2	83.7	78.5	85.0	90.0	88.0	87.0	85.0	83.0	40.1	84.3	85.8
Platinum	1,694	2,204	2,050	2,250	2,400	2,450	2,400	2,300	2,250	1,282	2,226	2,350
Palladium	1,478	1,706	1,490	1,550	1,600	1,550	1,500	1,450	1,400	1,151	1,586	1,475

Source: J.P. Morgan Commodities Research. *Base metals are LME cash and precious metals are spot.

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